

Design of a 15 kW Two-Stage Helical Gearbox for a Medium-Duty Belt Conveyor

Independent Machine Design Project
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1 Introduction

Gearboxes are widely used mechanical power transmission systems designed to modify speed and torque between a driving machine and a driven load. In many industrial applications, electric motors operate at relatively high rotational speeds, while the driven machinery requires lower speeds and higher torque. A gearbox provides this speed reduction while increasing the available output torque, allowing the system to operate within the required performance range.

This project focuses on the design of a two-stage helical gearbox for a medium-duty belt conveyor system. The gearbox is intended to transmit power from an electric motor operating at 1750 rpm and reduce the output speed to the range required by the conveyor drive. The selected input power is 15 kW, and the target output speed range is between 80 rpm and 100 rpm. To satisfy this requirement, a total speed reduction ratio of 20:1 is selected, resulting in an output speed of 87.5 rpm.

A two-stage parallel-shaft helical gearbox is selected for this application. The system consists of three shafts: an input shaft, an intermediate shaft, and an output shaft. The first gear stage reduces the speed from the input shaft to the intermediate shaft, while the second gear stage further reduces the speed from the intermediate shaft to the output shaft. Helical gears are preferred over spur gears due to their smoother operation, lower noise level, and better load-carrying capability under continuous industrial service.

The selected application is a medium-duty belt conveyor operating under low to moderate shock loading conditions. Such conveyors are commonly used for transporting packaged goods, agricultural products, or light bulk materials in industrial environments. The gearbox is designed with a gear and bearing life requirement of more than 12000 hours, while the shafts are designed for infinite fatigue life. These requirements make the project suitable for demonstrating fundamental machine design principles, including gear sizing, shaft design, bearing selection, key design, housing design, lubrication, and CAD assembly.

The main objective of this project is to develop a comprehensive and mechanically justifiable preliminary gearbox design. The design process includes defining the operating requirements, selecting the gear ratio, determining gear geometry, calculating gear forces, designing shafts for strength and fatigue life, selecting suitable bearings, and creating a complete CAD model of the gearbox assembly. The final design is intended to be suitable for engineering documentation, visualization, and portfolio presentation.

2 Application Scenario

The selected application for this gearbox is a medium-duty industrial belt conveyor system. Belt conveyors are widely used in manufacturing plants, warehouses, agricultural facilities, recycling lines, and bulk material handling systems. In such applications, an electric motor is commonly

used as the prime mover, while a gearbox is required to reduce the high motor speed to a lower conveyor drum speed and increase the available output torque.

In this project, the gearbox is assumed to drive a conveyor drum through an output shaft coupling. The conveyor operates under low to moderate shock loading conditions. This type of loading may occur due to starting and stopping of the conveyor, variation in transported material weight, and occasional uneven loading on the belt. However, the application is not considered to involve severe impact loads, sudden jamming, or highly irregular torque fluctuations.

The conveyor is assumed to transport packaged goods, agricultural products, or light bulk materials in an industrial environment. Therefore, the gearbox must provide reliable power transmission, sufficient output torque, acceptable service life, and a practical mechanical layout suitable for CAD modeling and assembly.

The selected input power is 15 kW, supplied by an electric motor operating at 1750 rpm. The required output speed range is between 80 rpm and 100 rpm. A total speed reduction ratio of 20:1 is selected, resulting in an output speed of 87.5 rpm. This output speed is suitable for driving a conveyor drum at a moderate belt speed.

For example, if a conveyor drum diameter of 250 mm is assumed, the belt speed can be estimated using

$$v = \frac{\pi D n}{60} \quad (1)$$

where v is the belt speed in m/s, D is the drum diameter in meters, and n is the drum rotational speed in rpm. For $D = 0.25$ m and $n = 87.5$ rpm,

$$v = \frac{\pi(0.25)(87.5)}{60} \quad (2)$$

$$v = 1.15 \text{ m/s} \quad (3)$$

This belt speed is reasonable for a medium-duty conveyor system, supporting the suitability of the selected output speed.

3 Design Requirements

The main design requirements of the gearbox are determined according to the selected conveyor application. The gearbox is required to transmit 15 kW of mechanical power from an electric motor

running at 1750 rpm and reduce the output speed to the required conveyor operating range. The selected gearbox ratio must produce an output speed between 80 rpm and 100 rpm.

The gearbox is designed for low to moderate shock loading conditions. The gears and bearings are required to have a service life greater than 12000 hours, while the shafts are designed for infinite fatigue life. The design must also be suitable for practical CAD modeling, assembly, and mechanical documentation.

Table 1: Main design requirements

Parameter	Value
Application	Medium-duty belt conveyor
Input power	15 kW
Input speed	1750 rpm
Required output speed	80–100 rpm
Selected total reduction ratio	20:1
Selected output speed	87.5 rpm
Number of stages	2
Number of shafts	3
Gear type	Helical gear
Loading condition	Low to moderate shock
Service factor	1.35
Design power	20.25 kW
First stage reduction ratio	4:1
Second stage reduction ratio	5:1
Normal pressure angle	20°
Helix angle	20°
Pinion profile shift coefficient	+0.3
Driven gear profile shift coefficient	−0.3
Gear material	AISI 4140 Q&T steel
Shaft material	AISI 4140 Q&T steel
Housing material	Gray cast iron
Required gear life	> 12000 h
Required bearing life	> 12000 h
Shaft design life	Infinite fatigue life
Coupling type	Flexible jaw coupling
Coupling location	Between motor and external journal bearing
Additional bearing type	External plain journal bearing
Plain bearing location	Before gearbox housing, on input shaft extension
Plain bearing material	SAE 660 bearing bronze
Plain bearing purpose	Auxiliary input shaft support

4 Gearbox Architecture

A two-stage parallel-shaft helical gearbox is selected for the conveyor drive application. The gearbox consists of three main shafts: the input shaft, the intermediate shaft, and the output shaft. The input shaft receives power from the electric motor through a flexible coupling and an external journal bearing support. Inside the gearbox, the first gear stage transmits power from the input shaft to the intermediate shaft, while the second gear stage transmits power from the intermediate shaft to the output shaft.

4.1 Selected Gearbox Type

The selected gearbox type is a two-stage parallel-shaft helical gear reducer. Helical gears are selected because they provide smoother meshing, lower noise, and higher load-carrying capability compared with spur gears. Since the application operates under low to moderate shock loading, a helical gearbox is suitable for reliable and continuous power transmission.

4.2 Power Transmission Layout

The complete power transmission path is arranged as follows:

$$\text{Motor} \rightarrow \text{Flexible Coupling} \rightarrow \text{Journal Bearing} \rightarrow \text{Gearbox} \rightarrow \text{Conveyor} \quad (4)$$

The flexible coupling is located between the electric motor and the external journal bearing support. The coupling transmits torque from the motor shaft to the input shaft extension while allowing small angular, parallel, and axial misalignments. A flexible jaw coupling is selected because of its simple construction, compact size, vibration damping capability, and suitability for low to moderate shock loading conditions.

An external plain journal bearing is placed on the input shaft extension before the gearbox housing. This bearing acts as an auxiliary support between the motor coupling and the gearbox. It is included to support the input shaft extension, reduce possible overhung loading, and demonstrate the design procedure of a sliding contact bearing. The internal gearbox shafts are still supported by rolling bearings.

4.3 Shaft Arrangement

The gearbox uses three shafts. The input shaft carries the first-stage pinion and receives power from the motor. The intermediate shaft carries both the first-stage driven gear and the second-stage

pinion. The output shaft carries the second-stage driven gear and delivers the reduced speed and increased torque to the conveyor drum.

Table 2: Shaft arrangement

Shaft	Function
Input shaft	Carries the first-stage pinion and receives power from the motor
Intermediate shaft	Carries the first-stage gear and second-stage pinion
Output shaft	Carries the second-stage gear and drives the conveyor

4.4 Reduction Distribution

The total selected speed reduction ratio is 20:1. This ratio is divided into two gear stages. The first stage has a reduction ratio of 4:1, while the second stage has a reduction ratio of 5:1.

$$i_{total} = i_1 i_2 \quad (5)$$

$$i_{total} = 4 \times 5 = 20 \quad (6)$$

For an input speed of 1750 rpm, the selected total reduction ratio gives an output speed of

$$n_{out} = \frac{1750}{20} \quad (7)$$

$$n_{out} = 87.5 \text{ rpm} \quad (8)$$

This output speed is within the required range of 80–100 rpm.

4.5 Profile Shift Strategy

Profile shift is applied to improve the tooth strength of the pinions and reduce the risk of undercutting. Since the pinions have fewer teeth than the driven gears, positive profile shift is assigned to the pinions. To keep the standard center distance unchanged, an equal negative profile shift is assigned to the mating gears. Therefore, the sum of profile shift coefficients for each gear pair is kept equal to zero.

For the first gear stage,

$$x_1 + x_2 = 0 \quad (9)$$

and for the second gear stage,

$$x_3 + x_4 = 0 \quad (10)$$

For both gear stages, the selected profile shift coefficients are

$$x_{pinion} = +0.3 \quad (11)$$

$$x_{gear} = -0.3 \quad (12)$$

This profile shift strategy strengthens the pinions while keeping the center distances unchanged.

5 Speed and Torque Calculations

The gearbox is designed to transmit power from a 15 kW electric motor operating at 1750 rpm. The selected total speed reduction ratio is 20:1, divided into two gear stages. The first stage has a reduction ratio of 4:1, and the second stage has a reduction ratio of 5:1.

The power and speed values are used to calculate the torque transmitted by each shaft. These torque values are later used in gear force analysis, shaft design, key design, bearing selection, and coupling selection.

5.1 Input Torque

The nominal input torque is calculated using

$$T = \frac{9550P}{n} \quad (13)$$

where T is the torque in Nm, P is the power in kW, and n is the rotational speed in rpm.

For the input shaft,

$$P = 15 \text{ kW} \quad (14)$$

$$n_{in} = 1750 \text{ rpm} \quad (15)$$

Therefore,

$$T_{in} = \frac{9550(15)}{1750} \quad (16)$$

$$T_{in} = 81.86 \text{ Nm} \quad (17)$$

Thus, the nominal torque transmitted by the input shaft is approximately

$$T_{in} \approx 82 \text{ Nm} \quad (18)$$

5.2 Shaft Speeds

The selected first-stage reduction ratio is

$$i_1 = 4 \quad (19)$$

and the selected second-stage reduction ratio is

$$i_2 = 5 \quad (20)$$

The intermediate shaft speed is obtained by dividing the input speed by the first-stage reduction ratio:

$$n_2 = \frac{n_1}{i_1} \quad (21)$$

$$n_2 = \frac{1750}{4} \quad (22)$$

$$n_2 = 437.5 \text{ rpm} \quad (23)$$

The output shaft speed is then calculated as

$$n_3 = \frac{n_2}{i_2} \quad (24)$$

$$n_3 = \frac{437.5}{5} \quad (25)$$

$$n_3 = 87.5 \text{ rpm} \quad (26)$$

The selected total ratio therefore provides an output speed within the required range of 80–100 rpm.

5.3 Efficiency Assumption

The gearbox contains two helical gear stages. For each gear stage, an efficiency of 97% is assumed. Therefore,

$$\eta_1 = 0.97 \quad (27)$$

$$\eta_2 = 0.97 \quad (28)$$

The total gearbox efficiency is

$$\eta_{total} = \eta_1 \eta_2 \quad (29)$$

$$\eta_{total} = 0.97 \times 0.97 \quad (30)$$

$$\eta_{total} = 0.9409 \quad (31)$$

Thus, the total gearbox efficiency is approximately 94.1%.

5.4 Shaft Torques

The nominal torque on the input shaft is

$$T_1 = 81.86 \text{ Nm} \quad (32)$$

The torque on the intermediate shaft is calculated by considering the first-stage reduction ratio and first-stage efficiency:

$$T_2 = T_1 i_1 \eta_1 \quad (33)$$

$$T_2 = 81.86(4)(0.97) \quad (34)$$

$$T_2 = 317.6 \text{ Nm} \quad (35)$$

The torque on the output shaft is calculated by considering the second-stage reduction ratio and second-stage efficiency:

$$T_3 = T_2 i_2 \eta_2 \quad (36)$$

$$T_3 = 317.6(5)(0.97) \quad (37)$$

$$T_3 = 1540.4 \text{ Nm} \quad (38)$$

Therefore, the gearbox increases the torque from approximately 82 Nm at the input shaft to approximately 1540 Nm at the output shaft.

Table 3: Shaft speeds and nominal torques

Shaft	Speed rpm	Nominal Torque Nm
Input shaft	1750	81.86
Intermediate shaft	437.5	317.6
Output shaft	87.5	1540.4

6 Service Factor and Design Power

The gearbox is designed for a medium-duty belt conveyor operating under low to moderate shock loading conditions. Since real machines are subjected to starting loads, small torque fluctuations, and possible uneven loading, a service factor is applied to the nominal power.

The selected service factor is

$$K_s = 1.35 \quad (39)$$

The design power is calculated as

$$P_d = K_s P \quad (40)$$

where P_d is the design power, K_s is the service factor, and P is the nominal input power.

$$P_d = 1.35(15) \quad (41)$$

$$P_d = 20.25 \text{ kW} \quad (42)$$

Therefore, the gearbox components are designed based on a design power of

$$P_d = 20.25 \text{ kW} \quad (43)$$

6.1 Design Torque Values

The nominal torque values are multiplied by the service factor to obtain the design torque values used in component sizing.

$$T_d = K_s T \quad (44)$$

For the input shaft,

$$T_{1d} = 1.35(81.86) \quad (45)$$

$$T_{1d} = 110.5 \text{ Nm} \quad (46)$$

For the intermediate shaft,

$$T_{2d} = 1.35(317.6) \quad (47)$$

$$T_{2d} = 428.8 \text{ Nm} \quad (48)$$

For the output shaft,

$$T_{3d} = 1.35(1540.4) \quad (49)$$

$$T_{3d} = 2079.5 \text{ Nm} \quad (50)$$

Table 4: Nominal and design torque values

Shaft	Nominal Torque Nm	Design Torque Nm
Input shaft	81.86	110.5
Intermediate shaft	317.6	428.8
Output shaft	1540.4	2079.5

7 Coupling Selection

A flexible coupling is used between the electric motor and the gearbox input shaft extension. The main function of the coupling is to transmit torque from the motor shaft to the gearbox input shaft while compensating for small shaft misalignments. In addition, the coupling helps reduce the transmission of vibration and shock loads between the motor and the gearbox.

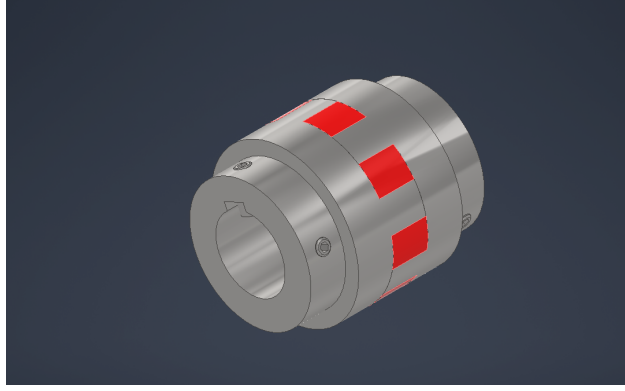


Figure 1: Selected coupling type modelled in CAD environment

In this design, a torsionally flexible jaw coupling is selected. This type of coupling consists of two metallic hubs and an elastomer spider. The elastomer element allows limited angular, parallel, and axial misalignment while providing torsional flexibility and damping. Therefore, a flexible jaw coupling is suitable for a medium-duty belt conveyor operating under low to moderate shock loading conditions.

The coupling is placed between the electric motor and the external plain journal bearing support. The general power transmission path is

$$\text{Motor} \rightarrow \text{Flexible Coupling} \rightarrow \text{External Journal Bearing} \rightarrow \text{Gearbox} \quad (51)$$

7.1 Working Torque

The coupling transmits the torque produced by the electric motor to the gearbox input shaft. During steady-state operation, the torque transmitted by the coupling is approximately equal to the working torque of the machine.

The nominal working torque is calculated from the motor power and speed:

$$M_{work} = \frac{9550P}{n} \quad (52)$$

where M_{work} is the working torque in Nm, P is the motor power in kW, and n is the motor speed in rpm.

For this design,

$$P = 15 \text{ kW} \quad (53)$$

$$n = 1750 \text{ rpm} \quad (54)$$

Therefore,

$$M_{work} = \frac{9550(15)}{1750} \quad (55)$$

$$M_{work} = 81.86 \text{ Nm} \quad (56)$$

Thus, the nominal working torque transmitted by the coupling is approximately

$$M_{work} \approx 82 \text{ Nm} \quad (57)$$

7.2 Coupling Design Torque

During steady-state operation, the coupling mainly transmits the working torque. However, during starting, stopping, and shock loading conditions, the coupling may be subjected to a higher torque. When the machine is started, the motor must provide both the working torque and the acceleration torque required to bring the rotating parts up to operating speed.

The motor starting torque requirement can be expressed as

$$M_{mot} = M_{work} + M_m \quad (58)$$

where M_m is the total acceleration torque required by the rotating system.

For the coupling, the required torque is associated with the machine side of the system. Therefore, the coupling must transmit the working torque and the acceleration torque of the rotating parts located after the coupling:

$$M_{KR} = M_{work} + M_{m2} \quad (59)$$

where M_{m2} is the acceleration torque of the machine-side rotating masses. In practical coupling selection, this effect is commonly represented by a coupling service factor. Therefore, the required coupling torque can be written as

$$M_{KR} = K_R M_{work} \quad (60)$$

where K_R is the coupling service factor. This factor accounts for the operating characteristics of the driven machine, starting conditions, and possible shock loads. For the present medium-duty belt conveyor application, a coupling service factor of

$$K_R = 1.35 \quad (61)$$

is selected.

Therefore, the required coupling design torque becomes

$$M_{KR} = 1.35(81.86) \quad (62)$$

$$M_{KR} = 110.5 \text{ Nm} \quad (63)$$

Thus, the selected coupling must satisfy

$$M_{coupling, rated} \geq M_{KR} \quad (64)$$

$$M_{coupling, rated} \geq 110.5 \text{ Nm} \quad (65)$$

7.3 Preliminary Coupling Selection

A torsionally flexible jaw coupling is selected based on the required coupling design torque, operating speed, and shaft bore requirement. The selected coupling should have a nominal rated torque higher than the calculated coupling design torque.

A preliminary selected coupling has a rated torque of

$$M_{coupling,rated} = 325 \text{ Nm} \quad (66)$$

The torque capacity check is therefore

$$325 > 110.5 \quad (67)$$

Since the rated torque of the selected coupling is greater than the required coupling design torque, the coupling is acceptable in terms of torque capacity.

The torque capacity ratio is

$$\frac{M_{coupling,rated}}{M_{KR}} = \frac{325}{110.5} \quad (68)$$

$$\frac{M_{coupling,rated}}{M_{KR}} = 2.94 \quad (69)$$

Therefore, the selected coupling has approximately 2.94 times the required design torque capacity.

7.4 Starting Torque Consideration

In addition to the service-factor-based design torque, the motor starting torque should also be considered. Electric motors may produce a starting torque higher than the nominal operating torque. If the motor starting torque is assumed to be approximately 2.5 times the nominal torque, then

$$M_{start} = 2.5M_{work} \quad (70)$$

$$M_{start} = 2.5(81.86) \quad (71)$$

$$M_{start} = 204.7 \text{ Nm} \quad (72)$$

The selected coupling rated torque is also greater than this estimated starting torque:

$$325 > 204.7 \quad (73)$$

Therefore, the selected coupling provides sufficient preliminary margin against the assumed starting torque condition.

7.5 Speed and Bore Requirements

The coupling must also be suitable for the operating speed of the motor. The motor speed is

$$n = 1750 \text{ rpm} \quad (74)$$

Therefore, the selected coupling must have a rated speed greater than the operating speed:

$$n_{coupling,rated} > 1750 \text{ rpm} \quad (75)$$

The coupling bore must also be compatible with the motor shaft and the gearbox input shaft extension. In this preliminary design, the gearbox-side shaft diameter is assumed to be

$$d_{shaft} = 40 \text{ mm} \quad (76)$$

Therefore, the gearbox-side coupling bore must satisfy

$$d_{bore,max} \geq 40 \text{ mm} \quad (77)$$

The final coupling model should be selected from a manufacturer catalogue by checking rated torque, maximum torque, rated speed, maximum bore diameter, and allowable misalignment values.

7.6 Torsional Vibration Consideration

A flexible coupling affects the dynamic behavior of the drive system because the elastomer element behaves like a torsional spring and damper between the motor and the gearbox input shaft. Therefore, the coupling is not only a torque-transmitting element, but also a component that influences torsional vibration and shock transmission.

Operation near a torsional critical speed may lead to increased vibration, noise, heat generation in the elastomer element, and reduced service life. A detailed torsional vibration analysis requires the mass moments of inertia of the motor, coupling, gears, shafts, and driven machine, as well as the torsional stiffness and damping properties of the coupling.

In this project, the coupling selection is treated as a preliminary catalogue-based selection. The coupling is checked for torque capacity, starting torque, operating speed, bore compatibility, and general suitability for low to moderate shock loading. A detailed torsional critical speed analysis is considered outside the scope of this design.

7.7 Coupling Selection Summary

Table 5: Flexible coupling selection summary

Parameter	Value
Coupling type	Torsionally flexible jaw coupling
Location	Between motor and external journal bearing
Motor power	15 kW
Motor speed	1750 rpm
Working torque, M_{work}	81.86 Nm
Coupling service factor, K_R	1.35
Required coupling torque, M_{KR}	110.5 Nm
Estimated starting torque, M_{start}	204.7 Nm
Selected rated coupling torque	325 Nm
Operating speed	1750 rpm
Preliminary gearbox-side bore	40 mm
Main function	Torque transmission and misalignment compensation

8 External Input Journal Bearing Design

An external plain journal bearing is placed on the input shaft extension before the gearbox housing. The bearing is located after the flexible coupling and before the shaft enters the gearbox. Its purpose is to support the input shaft extension and provide an additional bearing design example based on hydrodynamic radial journal bearing theory.

The internal gearbox shafts are supported by rolling bearings. Therefore, the external journal bearing is not used as the primary support for the internal gear forces. Instead, it acts as an auxiliary external support on the input side of the drive system.

Because this external bearing is included primarily to demonstrate the hydrodynamic journal-bearing design procedure, a preliminary radial design load of 1000 N is retained. This assumed load represents a reasonable combined allowance for the external shaft and coupling weight, minor

alignment loads, and installation disturbances. It is not derived from the internal gear-mesh reactions; a production design would determine the actual external support load from the complete motor–coupling–shaft arrangement.

8.1 Bearing Type and Design Assumptions

The selected bearing type is a hydrodynamic radial journal bearing. The journal is made from AISI 4140 Q&T steel, while the bearing bushing is selected as SAE 660 bearing bronze. Oil lubrication is selected in order to promote hydrodynamic film formation during steady-state operation.

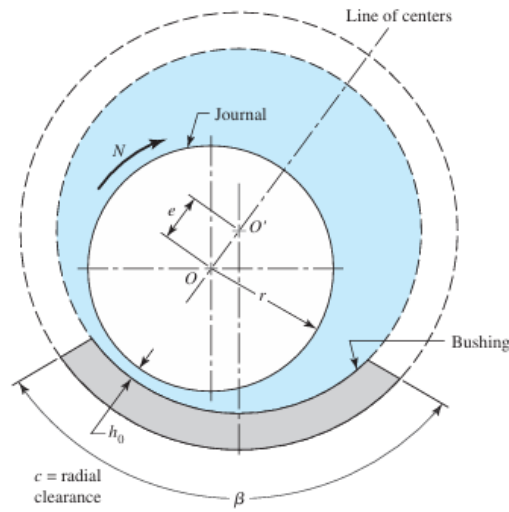


Figure 2: Bearing operation

The preliminary design assumptions are listed in Table 6.

8.2 Average Bearing Pressure

The average projected bearing pressure is calculated using

$$p_m = \frac{W}{ld} \quad (78)$$

where W is the radial load, l is the bearing length, and d is the journal diameter.

Using

Table 6: External journal bearing design assumptions

Parameter	Value
Bearing type	Hydrodynamic radial journal bearing
Journal material	AISI 4140 Q&T steel
Bearing material	SAE 660 bearing bronze
Journal diameter, d	40 mm
Journal radius, r	20 mm
Bearing length, l	40 mm
Length-to-diameter ratio, l/d	1
Radial clearance, δ_r	0.04 mm
Shaft speed, n	1750 rpm
Radial bearing load, W	1000 N
Selected lubricant	ISO VG 68 industrial gear oil
Lubricant compatibility	Bronze-compatible / yellow-metal compatible
Assumed oil-film operating temperature	Approximately 67°C
Dynamic viscosity, η	0.02 Pa·s
Lubrication method	Oil lubrication

$$W = 1000 \text{ N} \quad (79)$$

$$l = 0.04 \text{ m} \quad (80)$$

$$d = 0.04 \text{ m} \quad (81)$$

the average bearing pressure becomes

$$p_m = \frac{1000}{(0.04)(0.04)} \quad (82)$$

$$p_m = 625000 \text{ Pa} \quad (83)$$

$$p_m = 0.625 \text{ MPa} \quad (84)$$

8.3 Sommerfeld Number

The Sommerfeld number is used as the main dimensionless operating parameter for the hydrodynamic journal bearing. It is calculated as

$$S = \left(\frac{r}{\delta_r} \right)^2 \frac{\eta N}{p_m} \quad (85)$$

where r is the journal radius, δ_r is the radial clearance, η is the dynamic viscosity of the lubricant, N is the rotational speed in revolutions per second, and p_m is the average bearing pressure.

The shaft speed is

$$N = \frac{1750}{60} \quad (86)$$

$$N = 29.17 \text{ s}^{-1} \quad (87)$$

The clearance ratio term is

$$\frac{r}{\delta_r} = \frac{20}{0.04} \quad (88)$$

$$\frac{r}{\delta_r} = 500 \quad (89)$$

Therefore,

$$S = 500^2 \frac{(0.02)(29.17)}{625000} \quad (90)$$

$$S = 0.233 \quad (91)$$

Thus, the Sommerfeld number is approximately

$$S \approx 0.23 \quad (92)$$

Since the obtained Sommerfeld number is greater than the recommended minimum value of approximately 0.16 for normal surface quality, the bearing is considered suitable for hydrodynamic operation in this preliminary design.

8.4 Graphical Hydrodynamic Bearing Parameters

Using $S \approx 0.23$ and $l/d = 1$, the following dimensionless values are obtained from the hydrodynamic journal bearing design charts:

Table 7: Dimensionless parameters obtained from journal bearing charts

Parameter	Value
Relative minimum oil film thickness, $\delta = h_0/\delta_r$	0.56
Relative eccentricity, ε	0.44
Friction factor, K_f	5.25
Flow factor, K_q	4.05
Pressure ratio, p_m/p_{max}	0.47

The values of δ and ε are consistent with the relation

$$\delta = 1 - \varepsilon \quad (93)$$

$$0.56 = 1 - 0.44 \quad (94)$$

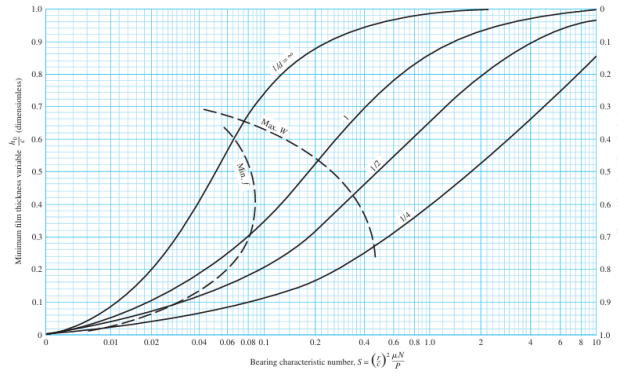


Figure 3: Chart for minimum film thickness variable and eccentricity ratio

8.5 Minimum Oil Film Thickness

The relative minimum oil film thickness is defined as

$$\delta = \frac{h_0}{\delta_r} \quad (95)$$

where h_0 is the minimum oil film thickness. From the chart,

$$\delta = 0.56 \quad (96)$$

Therefore,

$$h_0 = \delta \delta_r \quad (97)$$

$$h_0 = 0.56(0.04) \quad (98)$$

$$h_0 = 0.0224 \text{ mm} \quad (99)$$

$$h_0 = 22.4 \mu\text{m} \quad (100)$$

The minimum oil film thickness is therefore

$$h_0 = 22.4 \mu\text{m} \quad (101)$$

8.6 Surface Roughness Check

For safe hydrodynamic operation, the minimum oil film thickness should be greater than the sum of the surface roughness heights of the journal and bearing surfaces:

$$h_0 > R_{t1} + R_{t2} \quad (102)$$

For the journal and bronze bushing surfaces, the following surface roughness values are assumed:

$$R_{t1} = 4 \mu\text{m} \quad (103)$$

$$R_{t2} = 6 \mu\text{m} \quad (104)$$

Thus,

$$R_{t1} + R_{t2} = 10 \mu\text{m} \quad (105)$$

The surface roughness condition becomes

$$22.4 > 10 \quad (106)$$

Therefore, the minimum oil film thickness is greater than the combined roughness height of the two surfaces. This indicates that the bearing can maintain separation between the journal and bushing surfaces during steady-state hydrodynamic operation.

8.7 Maximum Oil Film Pressure

The pressure ratio obtained from the chart is

$$\frac{p_m}{p_{max}} = 0.47 \quad (107)$$

Therefore, the maximum oil film pressure is

$$p_{max} = \frac{p_m}{0.47} \quad (108)$$

$$p_{max} = \frac{0.625}{0.47} \quad (109)$$

$$p_{max} = 1.33 \text{ MPa} \quad (110)$$

The maximum oil film pressure is therefore approximately

$$p_{max} \approx 1.33 \text{ MPa} \quad (111)$$

8.8 Friction Coefficient and Power Loss

The friction factor obtained from the chart is

$$K_f = 5.25 \quad (112)$$

The friction factor is defined as

$$K_f = f \frac{r}{\delta_r} \quad (113)$$

Therefore, the coefficient of friction is

$$f = K_f \frac{\delta_r}{r} \quad (114)$$

$$f = 5.25 \frac{0.04}{20} \quad (115)$$

$$f = 0.0105 \quad (116)$$

The sliding velocity at the journal surface is

$$V = \frac{\pi d n}{60} \quad (117)$$

$$V = \frac{\pi(0.04)(1750)}{60} \quad (118)$$

$$V = 3.67 \text{ m/s} \quad (119)$$

The friction power loss is calculated as

$$P_f = f W V \quad (120)$$

$$P_f = (0.0105)(1000)(3.67) \quad (121)$$

$$P_f = 38.5 \text{ W} \quad (122)$$

Thus, the power loss due to friction in the journal bearing is approximately

$$P_f \approx 39 \text{ W} \quad (123)$$

8.9 Lubricant Flow Rate

The flow factor is defined as

$$K_q = \frac{q}{r\delta_r nl} \quad (124)$$

where q is the lubricant flow rate. Therefore,

$$q = K_q r \delta_r n l \quad (125)$$

Using

$$K_q = 4.05 \quad (126)$$

$$r = 0.02 \text{ m} \quad (127)$$

$$\delta_r = 0.00004 \text{ m} \quad (128)$$

$$n = 29.17 \text{ s}^{-1} \quad (129)$$

$$l = 0.04 \text{ m} \quad (130)$$

the lubricant flow rate is

$$q = 4.05(0.02)(0.00004)(29.17)(0.04) \quad (131)$$

$$q = 3.78 \times 10^{-6} \text{ m}^3/\text{s} \quad (132)$$

Converting to liters per minute,

$$q = 3.78 \times 10^{-6}(60000) \quad (133)$$

$$q = 0.227 \text{ L/min} \quad (134)$$

Therefore, the estimated chart-based hydrodynamic lubricant flow rate is approximately

$$q \approx 0.23 \text{ L/min} \quad (135)$$

8.10 Thermal Check

The mechanical power lost due to friction is assumed to be converted into heat. Therefore, the heat generation rate is approximately equal to the friction power loss:

$$\dot{Q}_{gen} \approx P_f \quad (136)$$

$$\dot{Q}_{gen} = 38.5 \text{ W} \quad (137)$$

The generated heat is assumed to be mainly carried away by the lubricant flow. The mass flow rate of the lubricant is

$$\dot{m} = \rho q \quad (138)$$

Assuming an oil density of

$$\rho = 850 \text{ kg/m}^3 \quad (139)$$

the mass flow rate becomes

$$\dot{m} = 850(3.78 \times 10^{-6}) \quad (140)$$

$$\dot{m} = 0.00321 \text{ kg/s} \quad (141)$$

The lubricant temperature rise is estimated using

$$\Delta T = \frac{\dot{Q}_{gen}}{\dot{m}c_p} \quad (142)$$

where c_p is the specific heat of the lubricant. Assuming

$$c_p = 2000 \text{ J/(kgK)} \quad (143)$$

the temperature rise is

$$\Delta T = \frac{38.5}{(0.00321)(2000)} \quad (144)$$

$$\Delta T = 6.0^\circ\text{C} \quad (145)$$

The calculated oil temperature rise is relatively low. Therefore, the bearing is considered acceptable from a preliminary thermal point of view.

Heat dissipation by natural convection from the bearing housing is neglected in this preliminary calculation. In practice, part of the generated heat would also be dissipated from the bearing housing to the surrounding air, further reducing the lubricant temperature rise.

8.11 Lubricant Selection

The lubricant viscosity used in the hydrodynamic journal bearing calculation is

$$\eta = 0.02 \text{ Pa} \cdot \text{s} \quad (146)$$

Therefore, the lubricant should have approximately this dynamic viscosity at the expected operating temperature of the bearing. A viscosity-temperature chart for ISO and SAE oils is used to select a suitable oil grade.

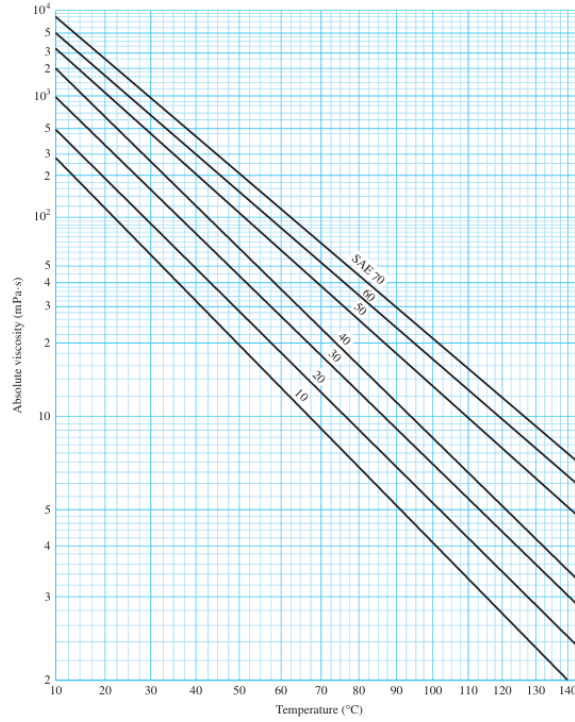


Figure 4: SAE Viscosity Chart

From the viscosity-temperature chart, the following approximate temperatures correspond to a dynamic viscosity of $0.02 \text{ Pa} \cdot \text{s}$:

Table 8: Oil grades corresponding to $\eta = 0.02 \text{ Pa} \cdot \text{s}$

Oil grade	Approximate temperature
ISO VG 68	67°C
ISO VG 100	75°C
ISO VG 150	85°C
SAE 10	50°C
SAE 30	65°C

For this design, ISO VG 68 is selected. At approximately 67°C, ISO VG 68 provides a dynamic viscosity close to

$$\eta = 0.02 \text{ Pa} \cdot \text{s} \quad (147)$$

which is consistent with the viscosity used in the Sommerfeld number calculation.

The same lubricant is selected for both the gearbox and the external journal bearing in order to simplify lubrication and maintenance. Since the journal bearing bushing is made of SAE 660 bearing bronze, the selected oil should be compatible with bronze and other yellow metals. Therefore, a bronze-compatible ISO VG 68 industrial gear oil is selected for the preliminary design.

The selected ISO VG 68 oil is also used as the common lubricant for the gearbox and the external journal bearing. This simplifies lubrication and maintenance while keeping the journal bearing calculation consistent with the selected lubricant viscosity.

8.12 Journal Bearing Design Summary

Table 9: External journal bearing design summary

Parameter	Value
Bearing type	Hydrodynamic radial journal bearing
Journal diameter, d	40 mm
Bearing length, l	40 mm
Length-to-diameter ratio, l/d	1
Radial clearance, δ_r	0.04 mm
Bearing load, W	1000 N
Shaft speed, n	1750 rpm
Selected lubricant	ISO VG 68 industrial gear oil
Assumed oil-film operating temperature	Approximately 67°C
Dynamic viscosity, η	0.02 Pa·s
Average bearing pressure, p_m	0.625 MPa
Sommerfeld number, S	0.23
Relative minimum film thickness, δ	0.56
Minimum oil film thickness, h_0	22.4 μm
Relative eccentricity, ε	0.44
Maximum oil film pressure, p_{max}	1.33 MPa
Coefficient of friction, f	0.0105
Friction power loss, P_f	38.5 W
Lubricant flow rate, q	0.23 L/min
Estimated oil temperature rise, ΔT	6.0°C

9 Gear Ratio and Tooth Number Selection

The gearbox is designed for an input power of 15 kW and an input speed of 1750 rpm. The required output speed is between 80 rpm and 100 rpm. The required total transmission ratio can therefore be estimated from

$$i_{\text{req}} = \frac{n_{\text{in}}}{n_{\text{out}}}. \quad (148)$$

A total transmission ratio of

$$i_{\text{tot}} = 20 \quad (149)$$

was selected. The resulting output speed is

$$n_{\text{out}} = \frac{n_{\text{in}}}{i_{\text{tot}}} \quad (150)$$

$$= \frac{1750}{20} \quad (151)$$

$$= 87.5 \text{ rpm}. \quad (152)$$

This value is within the required output-speed range.

Since a two-stage gearbox is used, the total ratio is divided into two individual stage ratios according to

$$i_{\text{tot}} = i_1 i_2. \quad (153)$$

The selected stage ratios are

$$i_1 = 4, \quad i_2 = 5. \quad (154)$$

Therefore,

$$i_{\text{tot}} = 4 \times 5 = 20. \quad (155)$$

The selected numbers of teeth are given in Table 10.

Table 10: Selected gear tooth numbers and stage ratios

Stage	Driving Gear	Driven Gear	Ratio
1	$z_1 = 20$	$z_2 = 80$	$i_1 = 4$
2	$z_3 = 20$	$z_4 = 100$	$i_2 = 5$

The first-stage ratio obtained from the selected tooth numbers is

$$i_1 = \frac{z_2}{z_1} = \frac{80}{20} = 4. \quad (156)$$

The second-stage ratio is

$$i_2 = \frac{z_4}{z_3} = \frac{100}{20} = 5. \quad (157)$$

Thus, the total transmission ratio is

$$i_{\text{tot}} = i_1 i_2 = 4 \times 5 = 20. \quad (158)$$

The corresponding shaft speeds are

$$n_1 = 1750 \text{ rpm}, \quad (159)$$

$$n_2 = \frac{n_1}{i_1} \quad (160)$$

$$= \frac{1750}{4} \quad (161)$$

$$= 437.5 \text{ rpm}, \quad (162)$$

and

$$n_3 = \frac{n_2}{i_2} \quad (163)$$

$$= \frac{437.5}{5} \quad (164)$$

$$= 87.5 \text{ rpm.} \quad (165)$$

A minimum actual tooth number of 20 was selected for both pinions. For a helical gear, the virtual number of teeth is calculated as

$$z_v = \frac{z}{\cos^3 \beta}. \quad (166)$$

For the pinions, with $z = 20$ and $\beta = 20^\circ$,

$$z_v = \frac{20}{\cos^3(20^\circ)} \quad (167)$$

$$\approx 24.1. \quad (168)$$

The virtual tooth number is greater than the commonly used minimum tooth number for a standard 20° involute gear. Therefore, severe undercutting is not expected. Nevertheless, a positive profile shift was applied to both pinions in order to increase the tooth-root thickness and improve their bending strength.

The selected profile-shift coefficients are

$$x_1 = +0.3, \quad x_2 = -0.3, \quad (169)$$

$$x_3 = +0.3, \quad x_4 = -0.3. \quad (170)$$

Thus, the total profile-shift coefficient of each gear pair is zero:

$$x_1 + x_2 = 0, \quad (171)$$

$$x_3 + x_4 = 0. \quad (172)$$

Since the total profile shift of each pair is zero, the theoretical center distance remains equal to the standard center distance. The positive shift strengthens the pinions, while the corresponding negative shift is assigned to the larger mating gears.

The actual radial profile-shift amount is obtained from

$$\Delta r = xm_n. \quad (173)$$

For the selected normal module of $m_n = 3$ mm, the profile-shift amounts are

$$\Delta r_{\text{pinion}} = (+0.3)(3) = +0.9 \text{ mm}, \quad (174)$$

and

$$\Delta r_{\text{gear}} = (-0.3)(3) = -0.9 \text{ mm}. \quad (175)$$

10 Gear Geometry Design

After selecting the gear ratios, tooth numbers, and profile-shift coefficients, the basic geometry of the two helical gear stages was determined. The same normal module, normal pressure angle, and helix angle were used for both stages.

The selected basic gear parameters are given in Table 11.

Table 11: Basic gear geometry parameters

Parameter	Value
Normal module, m_n	3 mm
Normal pressure angle, ϕ_n	20°
Helix angle, β	20°
Pinion profile-shift coefficient	+0.3
Gear profile-shift coefficient	-0.3
Stage 1 face width, b_1	40 mm
Stage 2 face width, b_2	45 mm

For a helical gear, the transverse module is calculated as

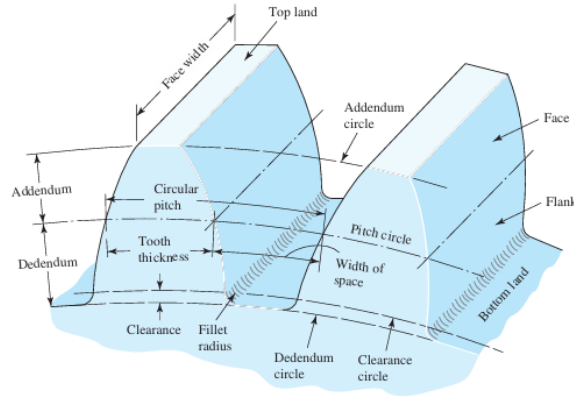


Figure 5: Basic nomenclature of an external helical gear pair

$$m_t = \frac{m_n}{\cos \beta}. \quad (176)$$

Substituting the selected values gives

$$m_t = \frac{3}{\cos(20^\circ)} \quad (177)$$

$$= 3.193 \text{ mm}. \quad (178)$$

The transverse pressure angle is obtained from

$$\tan \phi_t = \frac{\tan \phi_n}{\cos \beta}. \quad (179)$$

Therefore,

$$\phi_t = \tan^{-1} \left(\frac{\tan(20^\circ)}{\cos(20^\circ)} \right) \quad (180)$$

$$= 21.17^\circ. \quad (181)$$

The pitch diameter of each gear is calculated using

$$d = m_t z. \quad (182)$$

The pitch diameters are not directly changed by the profile-shift coefficients. Using the conventional full-depth tooth proportions, the addendum and dedendum heights are expressed as

$$h_a = m_n(1 + x), \quad (183)$$

and

$$h_f = m_n(1.25 - x). \quad (184)$$

Accordingly, the outside and root diameters are

$$d_a = d + 2m_n(1 + x), \quad (185)$$

and

$$d_f = d - 2m_n(1.25 - x). \quad (186)$$

For the positively shifted pinions, $x = +0.3$:

$$d_{a,\text{pinion}} = d + 7.8 \text{ mm}, \quad (187)$$

$$d_{f,\text{pinion}} = d - 5.7 \text{ mm}. \quad (188)$$

For the negatively shifted driven gears, $x = -0.3$:

$$d_{a,\text{gear}} = d + 4.2 \text{ mm}, \quad (189)$$

$$d_{f,\text{gear}} = d - 9.3 \text{ mm}. \quad (190)$$

The calculated gear diameters are listed in Table 12.

Table 12: Calculated profile-shifted gear diameters

Gear	Teeth	x	d [mm]	d_a [mm]	d_f [mm]	Face Width [mm]
Pinion 1	20	+0.3	63.851	71.651	58.151	40
Gear 1	80	−0.3	255.403	259.603	246.103	40
Pinion 2	20	+0.3	63.851	71.651	58.151	45
Gear 2	100	−0.3	319.253	323.453	309.953	45

Because the sum of the profile-shift coefficients is zero for each gear pair, the theoretical center distance is calculated from the pitch diameters in the usual manner:

$$a = \frac{d_p + d_g}{2}. \quad (191)$$

For the first stage,

$$a_1 = \frac{63.851 + 255.403}{2} \quad (192)$$

$$= 159.627 \text{ mm}. \quad (193)$$

For the second stage,

$$a_2 = \frac{63.851 + 319.253}{2} \quad (194)$$

$$= 191.552 \text{ mm}. \quad (195)$$

The center distances are summarized in Table 13.

Table 13: Gear-pair center distances

Stage	Pinion Pitch Diameter [mm]	Gear Pitch Diameter [mm]	Center Distance [mm]
1	63.851	255.403	159.627
2	63.851	319.253	191.552

The first-stage face width was selected as 40 mm. The second-stage face width was increased to 45 mm because the second stage transmits a substantially larger torque. The larger face width improves both the tooth bending strength and the contact-load capacity of the second-stage gears.

The hub dimensions used in the CAD model are summarized in Table 14.

Table 14: Gear and pinion dimensions used in the CAD model

Component	Tooth Face Width [mm]	Hub or Body Diameter [mm]	Additional Axial Length [mm]
Pinion 1	40	50	10
Gear 1	40	90	20
Pinion 2	45	50	20
Gear 2	45	125	20

The first-stage pinion was modeled integrally with the input shaft. The second-stage pinion was also treated as an integral shaft-pinion component in the final design. This arrangement avoids weakening the relatively small pinion root sections with keyways and reduces local stress concentration.

The two larger driven gears were modeled as separate components and connected to their shafts by standard parallel keys. Their hub dimensions were selected to provide sufficient material around the shaft bores and keyways while maintaining a practical assembly layout.

11 Gear Force Analysis

The forces generated at the helical gear meshes were calculated using the design torques, pitch diameters, normal pressure angle, and helix angle. These forces were subsequently used in the preliminary shaft calculations and rolling-bearing selection.

The selected normal pressure angle and helix angle are

$$\phi_n = 20^\circ, \quad (196)$$

$$\beta = 20^\circ. \quad (197)$$

The transverse pressure angle is calculated as

$$\phi_t = \tan^{-1} \left(\frac{\tan \phi_n}{\cos \beta} \right). \quad (198)$$

Substituting the selected values gives

$$\phi_t \approx 21.17^\circ. \quad (199)$$

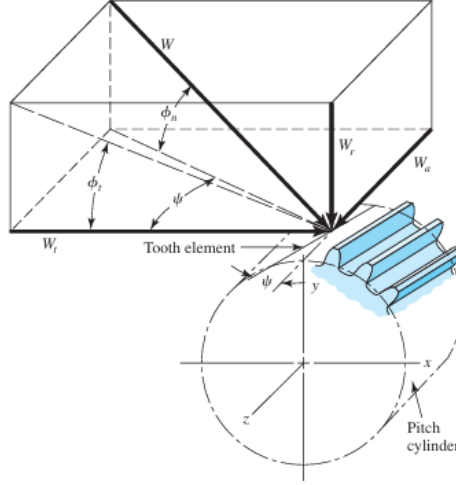


Figure 6: Tooth forces acting on a right-hand helical gear

For a helical gear, the tangential, radial, and axial force components are calculated using

$$F_t = \frac{2T}{d}, \quad (200)$$

$$F_r = F_t \tan \phi_t, \quad (201)$$

$$F_a = F_t \tan \beta, \quad (202)$$

where T is the transmitted torque and d is the pitch diameter of the pinion.

The normal tooth force is calculated as

$$F_n = \frac{F_t}{\cos \phi_n \cos \beta}. \quad (203)$$

11.1 First-Stage Gear Forces

The design torque acting on the first-stage input pinion is

$$T_{1,d} = 110.5 \text{ N m}. \quad (204)$$

The pitch diameter of the first-stage pinion is

$$d_1 = 63.851 \text{ mm.} \quad (205)$$

The tangential force is therefore calculated as

$$F_{t1} = \frac{2(110.5 \times 10^3)}{63.851} \quad (206)$$

$$= 3461.2 \text{ N.} \quad (207)$$

The radial force is

$$F_{r1} = F_{t1} \tan \phi_t \quad (208)$$

$$= 3461.2 \tan(21.17^\circ) \quad (209)$$

$$\approx 1340 \text{ N.} \quad (210)$$

The axial force is

$$F_{a1} = F_{t1} \tan \beta \quad (211)$$

$$= 3461.2 \tan(20^\circ) \quad (212)$$

$$= 1259.8 \text{ N.} \quad (213)$$

The corresponding normal tooth force is

$$F_{n1} = \frac{3461.2}{\cos(20^\circ) \cos(20^\circ)} \approx 3919 \text{ N.} \quad (214)$$

The pitch-line velocity of the first-stage pinion is calculated from

$$v_1 = \frac{\pi d_1 n_1}{60}, \quad (215)$$

where the pitch diameter is expressed in metres. Therefore,

$$v_1 \approx 5.85 \text{ m/s.} \quad (216)$$

The calculated first-stage gear forces are summarized in Table 15.

Table 15: First-stage gear forces

Force Component	Symbol	Value
Tangential force	F_{t1}	3461 N
Radial force	F_{r1}	1340 N
Axial force	F_{a1}	1260 N
Normal tooth force	F_{n1}	3919 N

11.2 Second-Stage Gear Forces

The second-stage pinion is mounted on the intermediate shaft. The design torque acting on the intermediate shaft is

$$T_{2,d} = 428.8 \text{ N m.} \quad (217)$$

The pitch diameter of the second-stage pinion is

$$d_3 = 63.851 \text{ mm.} \quad (218)$$

The tangential force is calculated as

$$F_{t2} = \frac{2(428.8 \times 10^3)}{63.851} \quad (219)$$

$$= 13429.7 \text{ N.} \quad (220)$$

The radial force is

$$F_{r2} = F_{t2} \tan \phi_t \quad (221)$$

$$= 13429.7 \tan(21.17^\circ) \quad (222)$$

$$= 5201.7 \text{ N.} \quad (223)$$

The axial force is

$$F_{a2} = F_{t2} \tan \beta \quad (224)$$

$$= 13429.7 \tan(20^\circ) \quad (225)$$

$$= 4888.0 \text{ N}. \quad (226)$$

The corresponding normal tooth force is

$$F_{n2} = \frac{13429.7}{\cos(20^\circ) \cos(20^\circ)} = 15208.8 \text{ N}. \quad (227)$$

The second-stage pitch-line velocity is

$$v_2 \approx 1.46 \text{ m/s}. \quad (228)$$

The calculated second-stage gear forces are summarized in Table 16.

Table 16: Second-stage gear forces

Force Component	Symbol	Value
Tangential force	F_{t2}	13430 N
Radial force	F_{r2}	5202 N
Axial force	F_{a2}	4888 N
Normal tooth force	F_{n2}	15209 N

11.3 Helix-Hand Selection and Shaft Axial Loads

The selected helix hands are shown in Table 17.

Table 17: Selected helix hands

Gear	Function	Helix Hand
Gear 1	First-stage input pinion	Right-hand
Gear 2	First-stage driven gear	Left-hand
Gear 3	Second-stage pinion	Left-hand
Gear 4	Second-stage output gear	Right-hand

The first-stage driven gear and the second-stage pinion are mounted on the intermediate shaft and have the same helix hand. However, the first-stage gear receives torque, whereas the second-stage

pinion transmits torque. Consequently, the axial forces generated by these two gears act in opposite directions on the intermediate shaft.

The net axial load acting on the intermediate shaft is therefore

$$F_{a,\text{int}} = F_{a2} - F_{a1} \quad (229)$$

$$= 4888.0 - 1259.8 \quad (230)$$

$$= 3628.2 \text{ N}. \quad (231)$$

The resulting design axial loads used in the bearing calculations are summarized in Table 18.

Table 18: Design axial loads acting on the shafts

Shaft	Design Axial Load
Input shaft	1260 N
Intermediate shaft	3628 N
Output shaft	4888 N

The selected helix-hand arrangement reduces the axial load acting on the intermediate-shaft bearings. If the two axial forces acted in the same direction, the resulting axial load would be

$$F_{a,\text{int,add}} = F_{a1} + F_{a2} \quad (232)$$

$$= 1259.8 + 4888.0 \quad (233)$$

$$= 6147.8 \text{ N}. \quad (234)$$

Therefore, the selected helix arrangement provides a more favourable bearing-loading condition by reducing the net axial force acting on the intermediate shaft.

12 Gear Strength Analysis

The gear strength calculations were carried out manually using the equations and correction factors presented in the selected machine-design reference. Autodesk Inventor was used only as an independent verification tool. The analytical calculations include tooth-root bending strength and surface contact pressure for both gear stages.

12.1 Calculation Factors

The service factor was already included in the design torques used to calculate the gear forces. Therefore, the operating factor in the following strength equations was taken as

$$K_0 = 1.00. \quad (235)$$

The dynamic factors were selected from the corresponding chart according to the pitch-line velocities and the assumed gear quality. The selected values were

$$K_{v1} = 1.06 \quad (236)$$

for the first stage and

$$K_{v2} = 1.02 \quad (237)$$

for the second stage. The load-distribution factor was selected as

$$K_m = 1.10. \quad (238)$$

For the positively profile-shifted pinions, the tooth-form factor was read from the reference chart using the virtual tooth number and the profile-shift coefficient. The virtual tooth number of each pinion is

$$z_v = \frac{z}{\cos^3 \beta} \quad (239)$$

and, for $z = 20$ and $\beta = 20^\circ$,

$$z_v \approx 24.1. \quad (240)$$

For $z_v \approx 24.1$ and $x = +0.3$, the selected tooth-form factor is

$$K_{fe} = 2.35. \quad (241)$$

12.2 Tooth-Root Bending Strength

The tooth-root bending stress was calculated from

$$\sigma_F = \frac{F_t}{bm_n} K_{fe} K_0 K_v K_m. \quad (242)$$

Detailed calculations were performed for the pinions because their lower tooth numbers and less favourable tooth geometry make them the critical members of the two gear pairs.

12.2.1 First-Stage Pinion

For the first-stage pinion,

$$F_{t1} = 3461.2 \text{ N}, \quad (243)$$

$$b_1 = 40 \text{ mm}, \quad (244)$$

$$m_n = 3 \text{ mm}, \quad (245)$$

$$K_{fe1} = 2.35, \quad (246)$$

$$K_0 = 1.00, \quad (247)$$

$$K_{v1} = 1.06, \quad (248)$$

$$K_m = 1.10. \quad (249)$$

Therefore,

$$\sigma_{F1} = \frac{3461.2}{40(3)} (2.35)(1.00)(1.06)(1.10) \quad (250)$$

$$\approx 79.0 \text{ MPa}. \quad (251)$$

12.2.2 Second-Stage Pinion

For the second-stage pinion,

$$F_{t2} = 13429.7 \text{ N}, \quad (252)$$

$$b_2 = 45 \text{ mm}, \quad (253)$$

$$m_n = 3 \text{ mm}, \quad (254)$$

$$K_{fe3} = 2.35, \quad (255)$$

$$K_0 = 1.00, \quad (256)$$

$$K_{v2} = 1.02, \quad (257)$$

$$K_m = 1.10. \quad (258)$$

Thus,

$$\sigma_{F3} = \frac{13429.7}{45(3)}(2.35)(1.00)(1.02)(1.10) \quad (259)$$

$$\approx 262.3 \text{ MPa}. \quad (260)$$

The second-stage pinion is therefore the more critical component in tooth-root bending.

12.3 Allowable Tooth-Root Strength

The ultimate tensile strength of the selected gear material was taken as

$$\sigma_u = 1570 \text{ MPa}. \quad (261)$$

The basic continuous tooth-root strength was estimated from

$$\sigma_D = 0.45\sigma_u. \quad (262)$$

Therefore,

$$\sigma_D = 0.45(1570) \quad (263)$$

$$= 706.5 \text{ MPa}. \quad (264)$$

The design life was selected as 12000 hours. The corresponding load-cycle numbers are in the long-life region; therefore, the life factor was taken as

$$K_L = 1.00. \quad (265)$$

For a reliability level of approximately 90%, the reliability factor was taken as

$$K_R = 0.897. \quad (266)$$

At the selected operating temperature, the temperature factor was taken as

$$K_T = 1.00. \quad (267)$$

The corrected continuous tooth-root strength is therefore

$$\sigma_D^* = K_L K_R K_T \sigma_D \quad (268)$$

$$= (1.00)(0.897)(1.00)(706.5) \quad (269)$$

$$\approx 633.7 \text{ MPa}. \quad (270)$$

The resulting tooth-root safety factors are

$$S_{F1} = \frac{\sigma_D^*}{\sigma_{F1}} = \frac{633.7}{79.0} \approx 8.02 \quad (271)$$

and

$$S_{F3} = \frac{\sigma_D^*}{\sigma_{F3}} = \frac{633.7}{262.3} \approx 2.42. \quad (272)$$

Both values are greater than the selected minimum bending safety requirement of 1.5. Therefore, both pinions are acceptable with respect to tooth-root bending strength.

12.4 Surface Contact Pressure

The maximum surface contact pressure was calculated from

$$p_H = K_E K_\alpha K_\beta K_i \sqrt{\frac{F_t}{bd} K_0 K_v K_m}. \quad (273)$$

Since both mating gears are made of steel, the elasticity factor was taken as

$$K_E = 189.8 \sqrt{\text{N/mm}^2}. \quad (274)$$

The contact-geometry and helix-angle factors were selected as

$$K_\alpha = 1.72, \quad (275)$$

$$K_\beta = 0.969. \quad (276)$$

The transmission-ratio factor was calculated from

$$K_i = \sqrt{\frac{i+1}{i}}. \quad (277)$$

For the first stage,

$$K_{i1} = \sqrt{\frac{4+1}{4}} = 1.118, \quad (278)$$

and for the second stage,

$$K_{i2} = \sqrt{\frac{5+1}{5}} = 1.095. \quad (279)$$

12.4.1 First-Stage Gear Pair

For the first stage,

$$p_{H1} = (189.8)(1.72)(0.969)(1.118) \sqrt{\frac{3461.2}{40(63.851)}} (1.00)(1.06)(1.10) \quad (280)$$

$$\approx 446 \text{ MPa}. \quad (281)$$

12.4.2 Second-Stage Gear Pair

For the second stage,

$$p_{H2} = (189.8)(1.72)(0.969)(1.095)\sqrt{\frac{13429.7}{45(63.851)}}(1.00)(1.02)(1.10) \quad (282)$$

$$\approx 795 \text{ MPa.} \quad (283)$$

The second-stage gear pair is therefore the governing condition for surface contact strength.

12.5 Allowable Surface Contact Strength

The basic continuous surface-contact strength was selected as

$$p_{HD} = 1288 \text{ MPa.} \quad (284)$$

The corrected continuous surface-contact strength was calculated from

$$p_{HD}^* = K_{L0}K_RK_Yp_{HD}. \quad (285)$$

For the selected design life, the surface life factor was taken as

$$K_{L0} = 1.00. \quad (286)$$

The reliability factor was taken as $K_R = 0.897$, and the lubrication factor for the selected ISO VG 68 oil was taken as

$$K_Y = 1.00. \quad (287)$$

Therefore,

$$p_{HD}^* = (1.00)(0.897)(1.00)(1288) \quad (288)$$

$$\approx 1155.3 \text{ MPa.} \quad (289)$$

The corresponding surface-contact safety factors are

$$S_{H1} = \frac{p_{HD}^*}{p_{H1}} = \frac{1155.3}{446} \approx 2.59 \quad (290)$$

and

$$S_{H2} = \frac{p_{HD}^*}{p_{H2}} = \frac{1155.3}{795} \approx 1.45. \quad (291)$$

Both stages satisfy the selected minimum surface-contact safety requirement of 1.3. However, the second-stage gear pair has the lower margin and governs the surface-strength design.

12.6 Analytical Strength Summary

Table 19: Summary of analytical gear-strength results

Component or Pair	Calculated Stress	Safety Factor	Result
First-stage pinion, bending	79.0 MPa	8.02	Acceptable
Second-stage pinion, bending	262.3 MPa	2.42	Acceptable
First-stage gear pair, contact	446 MPa	2.59	Acceptable
Second-stage gear pair, contact	795 MPa	1.45	Acceptable

The large driven gears were not evaluated separately in detailed tooth-root bending calculations because their substantially larger virtual tooth numbers provide more favourable tooth geometry and lower tooth-form factors than the corresponding pinions. Surface contact pressure was evaluated on a gear-pair basis.

12.7 Autodesk Inventor Verification

The manually calculated gear-strength results were compared with the ISO 6336:1996 results obtained from Autodesk Inventor. The software calculations were used only as an independent verification of the analytical design.

The Inventor results confirmed that the second-stage gear pair is the governing condition, particularly with respect to surface contact strength. The second-stage contact safety factors obtained from Inventor were approximately 1.22, while the manually calculated surface-contact safety factor was 1.45. The difference is attributed to the more detailed ISO 6336 correction factors included in the software model.

The manual calculation produced a second-stage contact safety factor of 1.45, slightly above the selected preliminary requirement of 1.3. Autodesk Inventor produced a lower value of approximately 1.22 because its ISO 6336 model includes more detailed correction factors. Therefore, the second-stage contact design is regarded as marginal at the preliminary-design level and remains the principal candidate for further optimization through increased face width or module, improved material or heat treatment, and refined load-distribution and lubrication assumptions.

13 Shaft Design and Verification

The gearbox contains three shafts operating under substantially different loading conditions. The input shaft is governed primarily by its rotational speed, the intermediate shaft carries the combined loading of both gear stages, and the output shaft transmits the highest torque. To avoid repeating the same derivation three times, the complete calculation procedure is presented for the intermediate shaft, which produced the lowest fatigue safety factor. The input and output shafts were evaluated by the same procedure and are summarized at the end of this section.

13.1 General Analysis Method

Gear forces were resolved into tangential, radial, and axial components. Bearing reactions were first obtained independently in the two mutually perpendicular bending planes. The resultant bending moment at a given shaft section was then calculated from

$$M_b = \sqrt{M_y^2 + M_z^2}. \quad (292)$$

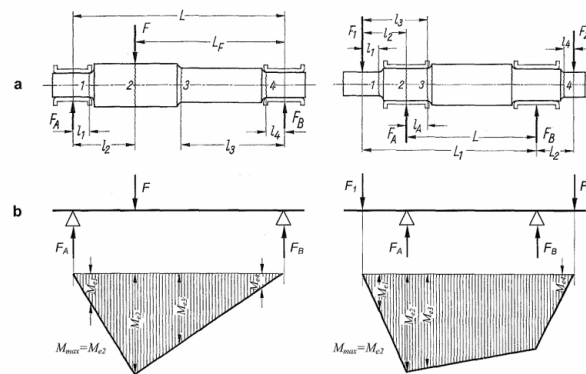


Figure 7: Representative free-body diagram used in the shaft reaction analysis

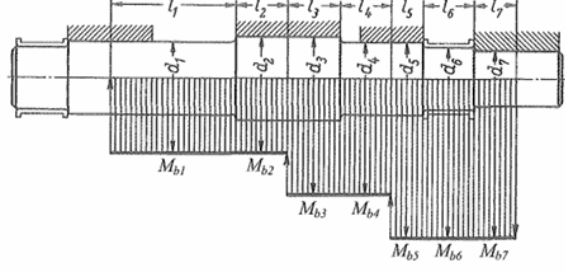


Figure 8: Shaft Torsional FBD example

For a solid circular shaft, the nominal bending, torsional, and axial stresses were calculated as

$$\sigma_b = \frac{32M_b}{\pi d^3}, \quad (293)$$

$$\tau_t = \frac{16T}{\pi d^3}, \quad (294)$$

$$\sigma_{ax} = \frac{4F_a}{\pi d^2}. \quad (295)$$

The static equivalent stress was evaluated using the von Mises criterion:

$$\sigma_{VM} = \sqrt{(\sigma_b + \sigma_{ax})^2 + 3\tau_t^2}. \quad (296)$$

The static safety factor was consequently obtained from

$$n_y = \frac{S_y}{\sigma_{VM}}. \quad (297)$$

St50 steel was selected for all shafts, with an assumed yield strength of $S_y = 295$ MPa and an ultimate tensile strength of $S_{ut} = 490$ MPa. The unmodified rotating-beam endurance limit was estimated as

$$S'_e = 0.5S_{ut} = 245 \text{ MPa}. \quad (298)$$

Because the shafts rotate under stationary transverse gear forces, the bending stress was treated as fully reversed. The transmitted torque and axial force were treated as steady mean loads. Therefore,

$$\sigma_a = \sigma_b, \quad \sigma_m = \sigma_{ax}, \quad \tau_a = 0, \quad \tau_m = \tau_t. \quad (299)$$

The corrected endurance limit was determined from

$$S_e = k_a k_b S'_e, \quad (300)$$

where the surface factor was taken as $k_a = 1.0$ and the size factor was selected according to the shaft diameter. The theoretical stress concentration factor was conservatively used as the fatigue stress concentration factor. The equivalent alternating and mean stresses were then calculated as

$$\sigma_{a,eq} = K_f \sigma_a, \quad (301)$$

$$\sigma_{m,eq} = \sqrt{(K_f \sigma_m)^2 + 3 (K_{fs} \tau_m)^2}. \quad (302)$$

Infinite-life fatigue safety was assessed using the modified Goodman relation:

$$\frac{\sigma_{a,eq}}{S_e} + \frac{\sigma_{m,eq}}{S_{ut}} = \frac{1}{n_f}. \quad (303)$$

Shaft deflection was evaluated using Euler–Bernoulli beam theory. For the constant-diameter approximation,

$$I = \frac{\pi d^4}{64}. \quad (304)$$

Critical speeds were estimated from the static-deflection method and, where multiple concentrated masses were present, by the Rayleigh equation:

$$\omega_{cr}^2 = g \frac{\sum W_i y_i}{\sum W_i y_i^2}, \quad n_{cr} = \frac{60 \omega_{cr}}{2\pi}. \quad (305)$$

13.2 Detailed Analysis of the Intermediate Shaft

The intermediate shaft operates at 437.5 rpm and transmits a design torque of 428.8 N m. The first-stage driven gear and the second-stage pinion are positioned between the bearings. Their locations, measured from bearing A, are

$$x_{G2} = 62.5 \text{ mm}, \quad x_{G3} = 321.25 \text{ mm}, \quad (306)$$

with a total bearing span of

$$L_{AB} = 396.25 \text{ mm}. \quad (307)$$

The tangential and radial forces acting on the two gears are oppositely directed. The applied gear forces are summarized in Table 20.

Table 20: Gear forces acting on the intermediate shaft.

Gear	F_t (N)	F_r (N)	F_a (N)
Gear 2	3461.2	1340.0	1259.8
Gear 3	13429.7	5201.7	4888.0

The opposing helix-force directions reduce the net axial load to

$$F_{a,\text{net}} = 4888.0 - 1259.8 = 3628.2 \text{ N.} \quad (308)$$

Solving the equilibrium equations independently in the tangential and radial planes gives

$$R_{At} = -373.4 \text{ N}, \quad R_{Bt} = 10341.9 \text{ N}, \quad (309)$$

$$R_{Ar} = -144.1 \text{ N}, \quad R_{Br} = 4005.8 \text{ N.} \quad (310)$$

The negative signs indicate that the actual reactions at bearing A act opposite to the initially assumed directions. The resultant radial reactions are

$$R_A = 0.400 \text{ kN}, \quad R_B = 11.09 \text{ kN.} \quad (311)$$

The maximum combined bending moment occurs near Gear 3. The bending components at this location are

$$M_y = 775.6 \text{ N m}, \quad M_z = 300.4 \text{ N m}, \quad (312)$$

which give

$$M_{b,\text{max}} = \sqrt{775.6^2 + 300.4^2} = 831.8 \text{ N m.} \quad (313)$$

For the 50 mm critical shaft diameter, the nominal stresses are

$$\sigma_b = 67.8 \text{ MPa}, \quad (314)$$

$$\tau_t = 17.47 \text{ MPa}, \quad (315)$$

$$\sigma_{ax} = 1.85 \text{ MPa.} \quad (316)$$

The corresponding von Mises stress and static safety factor are

$$\sigma_{\text{VM}} = 75.9 \text{ MPa}, \quad n_y = \frac{295}{75.9} = 3.89. \quad (317)$$

For the fatigue calculation, $k_a = 1.0$, $k_b = 0.82$, and $K_f = K_{fs} = 1.65$ were used. Thus,

$$S_e = (1.0)(0.82)(245) = 200.9 \text{ MPa.} \quad (318)$$

The equivalent alternating and mean stresses are

$$\sigma_{a,\text{eq}} = 1.65(67.8) = 111.9 \text{ MPa}, \quad (319)$$

$$\sigma_{m,\text{eq}} = \sqrt{[1.65(1.85)]^2 + 3[1.65(17.47)]^2} = 50.0 \text{ MPa.} \quad (320)$$

The modified Goodman safety factor is therefore

$$n_f = \left(\frac{111.9}{200.9} + \frac{50.0}{490} \right)^{-1} = 1.52. \quad (321)$$

A constant-diameter beam approximation gives a maximum operating deflection of approximately

$$\delta_{\max} \approx 0.128 \text{ mm.} \quad (322)$$

This value is conservative because the actual shaft contains locally larger diameters. It should nevertheless be considered when evaluating gear misalignment and face-load distribution.

The rotating masses used for the critical-speed estimate were

$$m_{G2} = 15.761 \text{ kg}, \quad m_{G3} = 0.438 \text{ kg.} \quad (323)$$

The corresponding gravity-induced deflections at the gear locations were approximately 0.000898 mm and 0.000730 mm. The Rayleigh method gives

$$n_{cr, \text{Rayleigh}} \approx 31630 \text{ rpm,} \quad (324)$$

while the conservative Dunkerley estimate is approximately

$$n_{cr, \text{Dunkerley}} \approx 31340 \text{ rpm.} \quad (325)$$

Both values are far above the operating speed of 437.5 rpm. Consequently, resonance is not a governing design condition for the intermediate shaft.

13.3 Summary of All Shaft Results

The input and output shafts were checked using the same static, fatigue, deflection, and critical-speed procedures. Since neither shaft governed the design, their full derivations are omitted to avoid unnecessary repetition. The principal results are summarized in Table 21.

Table 21: Summary of shaft verification results.

Parameter	Input shaft	Intermediate shaft	Output shaft
Shaft diameter (mm)	50	50	70
Operating speed (rpm)	1750	437.5	87.5
Design torque (N m)	110.5	428.8	2079.5
Maximum bending moment (N m)	117.8	831.8	532.9
von Mises stress (MPa)	12.87	75.9	56.15
Static safety factor, n_y	22.9	3.89	5.25
Fatigue size factor, k_b	0.82	0.82	0.79
Stress concentration factor	1.65	1.65	1.40
Fatigue safety factor, n_f	9.50	1.52	3.74
Maximum operating deflection (mm)	0.00246	0.128	0.00393
Estimated critical speed (rpm)	10960	31340–31630	103500

The input shaft has large static and fatigue margins because of its relatively low transmitted torque and short bearing span. Its reported critical-speed estimate includes the assumed 460 mm overhung coupling location and a constant 50 mm equivalent shaft diameter.

Although the output shaft transmits the highest torque, its 70 mm diameter provides adequate static and infinite-life fatigue resistance. Its operating speed is also extremely small compared with its estimated first critical speed.

The intermediate shaft is the governing member because it carries the loads from both gear stages and develops the lowest infinite-life fatigue safety factor. Nevertheless, the calculated value of $n_f = 1.52$ remains above unity, while its operating speed is far below the first critical-speed range.

14 Rolling Bearing Selection and Arrangement

The bearing selection was performed after the shaft reactions had been determined. Because the gearbox uses helical gears, each shaft is subjected to both radial and axial loading. Single-row angular-contact ball bearings were therefore selected and arranged in opposed pairs. The preliminary CAD model contained the initial bearing selections; however, the intermediate-shaft bearing was revised in the final design calculation because the conservative manual reaction exceeded the value used during the CAD pre-design stage.

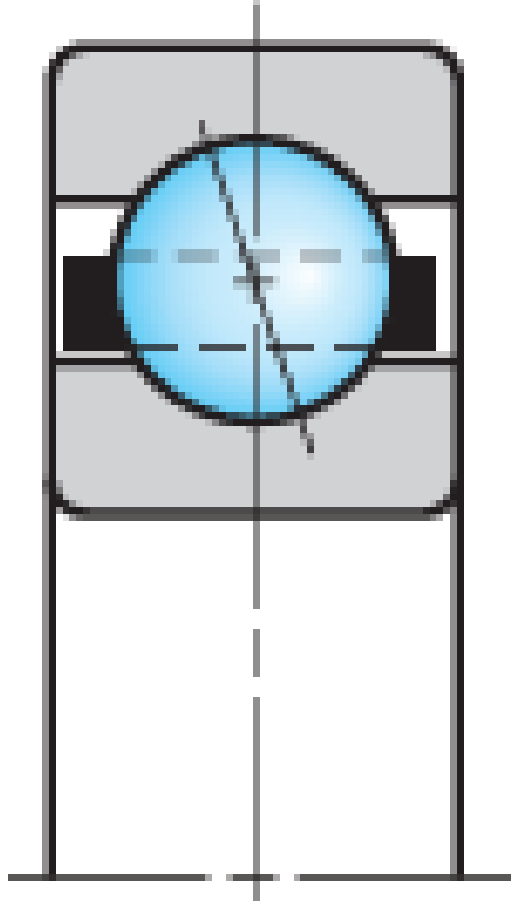


Figure 9: Single-row angular-contact ball bearing selected for the gearbox shafts

14.1 Bearing-Life Criterion

The basic rating life of a ball bearing is

$$L_{10} = \left(\frac{C}{P} \right)^3, \quad (326)$$

where C is the basic dynamic load rating and P is the equivalent dynamic bearing load. In operating hours,

$$L_{10h} = \frac{10^6}{60n} \left(\frac{C}{P} \right)^3. \quad (327)$$

The minimum required bearing life was specified as

$$L_{10h, \text{req}} = 12000 \text{ h}. \quad (328)$$

For preliminary capacity sizing, the equivalent load was first approximated by the resultant radial reaction,

$$P \approx F_r. \quad (329)$$

The corresponding required dynamic rating is

$$C_{\text{req}} = P \left(\frac{60nL_{10h,\text{req}}}{10^6} \right)^{1/3}. \quad (330)$$

The axial loads were considered when selecting the opposed angular-contact arrangement and its thrust direction. In each bearing pair, the bearing oriented to react the applied axial-force direction carries the dominant axial component. For the intermediate shaft, the arrangement was selected so that the net axial load is reacted primarily by the bearing having the lower radial reaction. However, the numerical life values reported below remain preliminary baseline values based on $P \approx F_r$. Final production verification must use the manufacturer-specific factors X , Y , and e in $P = XF_r + YF_a$, together with the actual preload, internal clearance, and load sharing of the opposed pair.

14.2 Preliminary Capacity Requirements

The radial reactions obtained from the shaft analysis were used to estimate the minimum required dynamic capacities. The critical manually calculated reaction occurred at bearing B of the intermediate shaft.

Table 22: Preliminary bearing-capacity requirements based on radial reactions.

Bearing position	F_r (kN)	Speed (rpm)	C_{req} (kN)
Input shaft A/B	1.856	1750.0	20.0
Intermediate shaft A	2.15	437.5	14.6
Intermediate shaft B	11.25	437.5	76.5
Output shaft A/B	7.20	87.5	28.6

The initial CAD model used an SKF 7309 bearing on the intermediate shaft. The earlier pre-design calculation had used a radial load of 8.53 kN, which resulted in a calculated life of approximately 13900 h. However, the conservative manual shaft-reaction analysis produced a higher critical radial load of approximately 11.25 kN. Therefore, the final bearing selection was increased by one series size rather than retaining the less conservative pre-design value.

14.3 Final Bearing Selections

The final bearing selections are summarized in Table 23. The intermediate shaft was upgraded to two SKF 7311 BECBP bearings to maintain a symmetric opposed pair and to provide adequate

capacity under the conservative reaction.

Table 23: Final bearing selections.

Shaft	Bearing	Quantity	Bore (mm)	Outside diameter (mm)	Width (mm)	C (kN)
Input	SKF 7308 BEP	2	40	90	23	46.2
Intermediate	SKF 7311 BECBP	2	55	120	29	85.0
Output	SKF 7213 BEP	2	65	120	23	66.3

Using the radial reactions as the baseline equivalent loads, the calculated rating lives are

$$L_{10h,\text{input}} = \frac{10^6}{60(1750)} \left(\frac{46.2}{1.856} \right)^3 \approx 1.47 \times 10^5 \text{ h}, \quad (331)$$

$$L_{10h,\text{intermediate,B}} = \frac{10^6}{60(437.5)} \left(\frac{85.0}{11.25} \right)^3 \approx 1.64 \times 10^4 \text{ h}, \quad (332)$$

$$L_{10h,\text{output}} = \frac{10^6}{60(87.5)} \left(\frac{66.3}{7.20} \right)^3 \approx 1.49 \times 10^5 \text{ h}. \quad (333)$$

The intermediate-shaft bearing remains the governing bearing. Its preliminary radial-load-based baseline rating life exceeds the required 12000 h, while the input and output bearings retain substantially larger baseline margins. A final catalogue-based combined-load verification using the manufacturer-specific equivalent dynamic load factors is required before production.

14.4 Opposed Angular-Contact Arrangement

Each shaft uses two single-row angular-contact ball bearings installed in a back-to-back, or O, arrangement. In this configuration, the contact lines diverge outward. Compared with a face-to-face arrangement, the effective support span is larger and the bearing pair provides greater resistance to overturning moments and shaft tilting. This is advantageous for maintaining gear alignment under combined radial and axial loading.

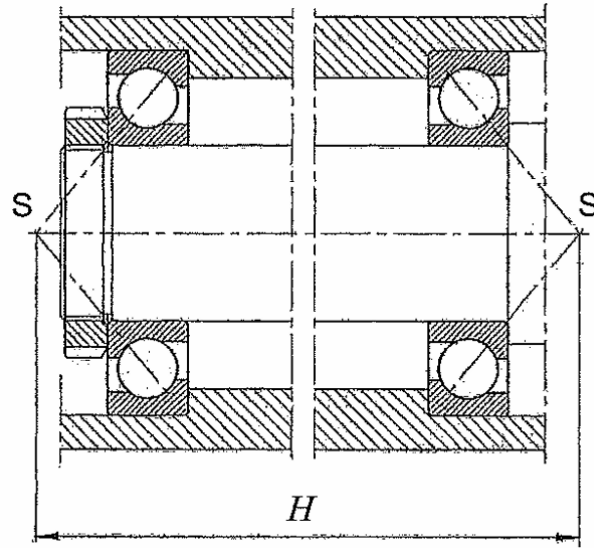


Figure 10: Back-to-back angular-contact bearing arrangement used for the gearbox shafts

The selected arrangements are:

- input shaft: 2× SKF 7308 BEP, back-to-back;
- intermediate shaft: 2× SKF 7311 BECBP, back-to-back;
- output shaft: 2× SKF 7213 BEP, back-to-back.

The helical-gear axial forces are reacted according to their direction. The input shaft is subjected to approximately 1.260 kN of axial force, the intermediate shaft to a net axial force of approximately 3.628 kN, and the output shaft to approximately 4.888 kN. Within each opposed pair, the bearing facing the applied thrust direction carries the dominant axial component, while the other bearing reacts thrust in the opposite direction.

For the intermediate shaft, the higher radial reaction occurs at bearing B. Therefore, the pair orientation and axial adjustment should be selected so that the net axial force is carried primarily by bearing A, which has the lower radial reaction. This reduces the equivalent dynamic load acting on the life-critical bearing B.

14.5 Axial Location, Adjustment, and Thermal Expansion

The two angular-contact bearings on each shaft act as an adjusted bearing pair rather than as a conventional independent locating and floating bearing combination. The inner rings rotate with

the shaft and are axially located by shaft shoulders, spacers, locknuts, or retaining rings. The outer rings are located by housing shoulders and end covers.

A controlled axial clearance or light preload should be established using cover adjustment or shims. Excessive preload must be avoided because thermal expansion of the shaft can increase the internal axial load, friction, and operating temperature. The final assembly should therefore provide sufficient adjustment to remove excessive play while still accommodating the expected thermal growth.

14.6 CAD Revision Note

The existing CAD model represents the preliminary bearing layout and was not revised after the conservative manual bearing calculation. The final intermediate-shaft selection was changed from the preliminary SKF 7309 bearing to the SKF 7311 BECBP. Consequently, a production-ready CAD revision would require corresponding changes to the shaft bearing seats, housing bores, shoulders, end covers, spacers, and retaining features. These changes are identified as final-detail modifications and do not alter the gearbox operating principle or the completed analytical verification.

14.7 Bearing-Selection Summary

The bearing design is governed by the high radial reaction at the intermediate-shaft B support. The preliminary 7309 bearing was adequate only for the lower CAD pre-design load and did not satisfy the required life when the more conservative manual reaction was used. Upgrading the intermediate-shaft pair to SKF 7311 BECBP bearings increased the basic dynamic rating sufficiently for the preliminary radial-load-based life to exceed the 12000 h target. Final manufacturer-specific verification under combined radial and axial loading remains necessary. All three shafts use back-to-back angular-contact pairs to provide bidirectional axial-load capacity and high moment stiffness.

15 Shaft–Hub Connections

Torque transmission between the shafts and the separately mounted gears is provided by parallel keys. The two pinions were not mounted as separate hubbed components. Because of their relatively small diameters, they were manufactured integrally with the corresponding shafts. This reduces the number of parts, eliminates a local keyway at the pinion location, improves concentricity, and avoids weakening an already compact high-load region.

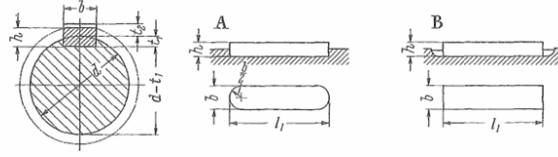


Figure 11: Key Connection

For a parallel key transmitting torque T through a shaft of diameter d , the tangential force acting on the key is

$$F_k = \frac{2T}{d}. \quad (334)$$

The key is checked against shear and bearing failure using

$$\tau_k = \frac{F_k}{bL}, \quad (335)$$

$$\sigma_{b,k} = \frac{2F_k}{hL}, \quad (336)$$

where b , h , and L are the key width, height, and effective length, respectively.

The output-side gear connection was selected as the detailed verification case because it transmits the maximum gearbox torque. The design torque is

$$T = 2079.5 \text{ N m} = 2.0795 \times 10^6 \text{ N mm}, \quad (337)$$

and the shaft diameter at the key seat is

$$d = 70 \text{ mm}. \quad (338)$$

The tangential force acting on the key is therefore

$$F_k = \frac{2(2.0795 \times 10^6)}{70} = 5.941 \times 10^4 \text{ N}. \quad (339)$$

Using the selected standard key dimensions and the allowable shear and bearing stresses, the governing minimum effective key length was found to be approximately

$$L_{\min} \approx 36 \text{ mm}. \quad (340)$$

A standard key length of

$$L = 40 \text{ mm} \quad (341)$$

was therefore selected. The remaining keyed connections were checked using the same procedure and were less critical because they transmit lower torques.

16 Shaft Shoulders, Retaining Rings, and Spacers

Axial positioning of gears, bearing inner rings, and spacers is provided by shaft shoulders, DIN 471 external retaining rings, and dedicated spacer components. A nominal shoulder height of

$$h_s = 2 \text{ mm} \quad (342)$$

was used at the relevant shaft diameter transitions.

The shaft shoulders perform two functions. First, they provide positive axial location for bearings and mounted components. Second, they introduce local changes in section geometry. These diameter transitions affect both bending and torsional strength because the local section modulus and polar section modulus change with diameter. In addition, a shoulder produces a stress concentration whose severity depends on the diameter ratio and fillet radius. The local peak stress may be represented approximately by

$$\sigma_{\max} = K_{t,b} \sigma_{\text{nom}}, \quad (343)$$

$$\tau_{\max} = K_{t,t} \tau_{\text{nom}}, \quad (344)$$

where $K_{t,b}$ and $K_{t,t}$ are the bending and torsional stress concentration factors. For fatigue calculations, the corresponding fatigue stress concentration factors were used conservatively in the shaft verification.

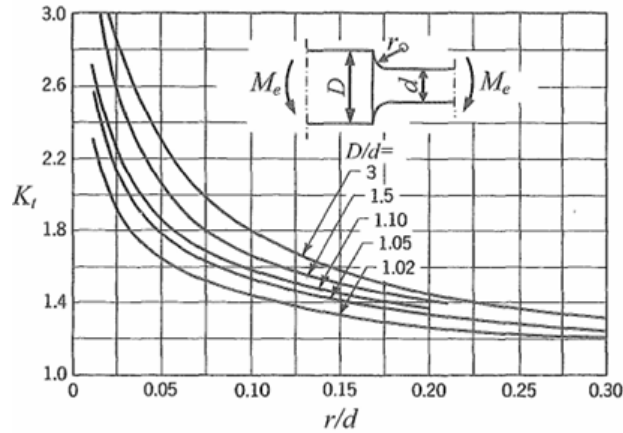


Figure 12: Stress Concentration Factors for a stepped shaft in bending

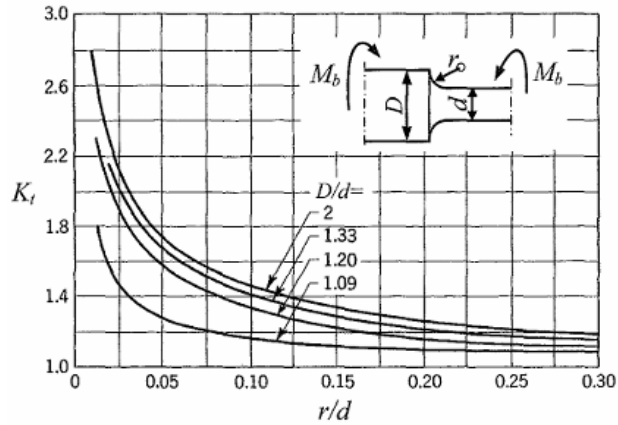


Figure 13: Stress Concentration Factors for a stepped shaft in torsion

DIN 471 external retaining rings were selected for the applicable shaft diameters. The associated groove dimensions were based on the relevant standard geometry. Because the retaining-ring grooves are standardized and contain a sharp local reduction in section, they generated high localized stresses in the finite-element analysis. These peaks were interpreted as stress-concentration effects rather than as representative stresses over a large shaft volume. Nevertheless, they confirm that retaining-ring grooves are critical geometric details and should be located away from sections carrying the maximum combined bending and torsional loading whenever possible.

Spacers were used between bearings, gears, and shoulders to maintain the required axial distances and to support the adjusted angular-contact bearing arrangements. These components transmit assembly preload and axial positioning forces, but they are not intended to carry the primary shaft torque. Their end faces should be flat and parallel to prevent uneven bearing-ring loading.

17 Sealing Elements and Lubricant Selection

Different sealing arrangements were used according to whether the shaft passed through the housing wall. The input and output shafts extend outside the gearbox; therefore, radial shaft seals were selected at these locations. Seal sizes are chosen to match the corresponding shaft diameters, and the running surfaces should have suitable finish, hardness, and assembly chamfers.

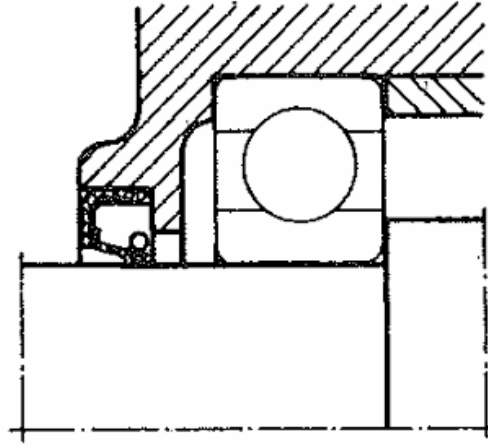


Figure 14: Contact sealing element

The intermediate shaft does not extend through the housing. For this reason, a contacting lip seal was unnecessary. A non-contact sealing arrangement was used instead, with a nominal radial clearance of

$$c = 0.2 \text{ mm.} \quad (345)$$

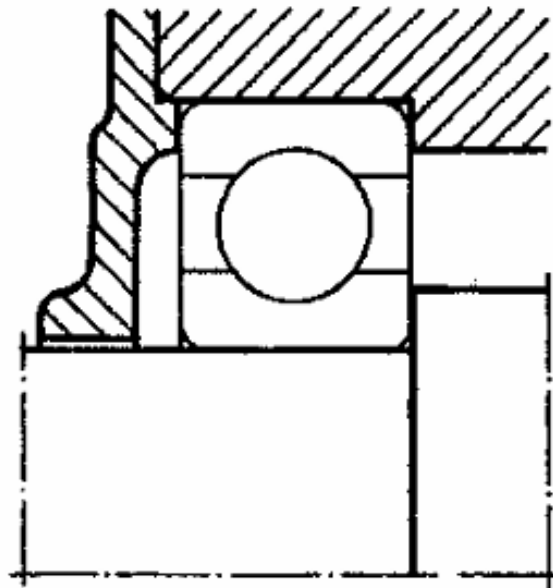


Figure 15: Contactless sealing element

The non-contact geometry forms a tortuous leakage path while avoiding sliding friction, heat genera-

tion, and seal wear. This is particularly appropriate for an internal shaft whose primary requirement is to restrict oil migration rather than to seal a housing penetration.

The same lubricant previously selected for the hydrodynamic journal bearing was adopted for the gearbox in order to simplify lubricant management. ISO VG 68 oil was used as the baseline selection. The lubricant collects in the lower region of the gearbox housing and forms an oil sump. The lower gear elements partially dip into this oil and distribute it by splash to the remaining gears and rolling bearings.

The oil level should be high enough to ensure reliable lubricant pickup, but not so high that excessive churning losses and heat generation occur. The housing should therefore include the following service elements:

- an oil filling port,
- a drain plug at the lowest practical point,
- an oil-level sight glass or observation unit,
- a breather to prevent internal pressure buildup,
- and, where practical, an inspection opening for visual examination.

The use of one common lubricant for the journal bearing, gears, and rolling bearings simplifies maintenance and reduces the risk of using incompatible fluids. The final oil quantity and operating level should be confirmed during commissioning by monitoring temperature, leakage, and churning behaviour.

18 Housing Design

The gearbox housing was developed as a rigid enclosure for the three-shaft, two-stage transmission. The principal external dimensions of the housing are approximately

$$390 \times 560 \times 410 \text{ mm.} \quad (346)$$

These dimensions were selected to provide sufficient internal clearance for the gears, shafts, bearings, lubrication volume, and assembly operations while keeping the overall gearbox compact.

The housing consists of two main casing halves. An internal bearing-support wall is incorporated into the structure to provide additional support for the shaft bearings and to reduce deformation of the bearing seats. This wall also helps maintain the relative alignment of the three shaft axes under operating loads. Because gear-mesh quality is sensitive to shaft misalignment, the stiffness of the bearing-support regions is more important than the stiffness of the unloaded housing panels.

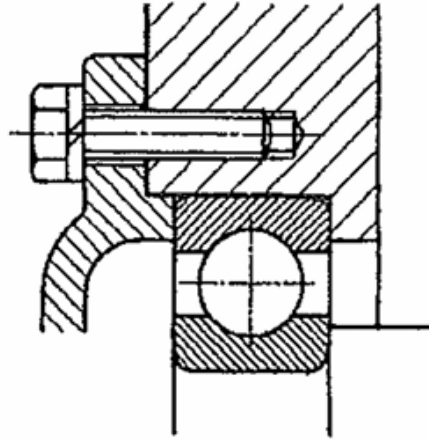


Figure 16: Bearing insertion example

External bearing covers are mounted at the shaft ends. These covers provide axial retention for the bearing outer rings, support the sealing elements, and allow access during assembly and maintenance. Where required, cover thicknesses and local bosses should be increased around the bearing seats to limit local deformation.

The two principal housing halves are fastened using

$$10 \times M8 \times 45 \quad (347)$$

bolts with matching nuts. The fasteners are distributed around the housing flange to maintain joint compression and preserve sealing along the mating surface. In a production design, suitable washers, locking features, and a gasket or liquid sealant would also be specified.

The CAD model was intentionally simplified to keep the project scope manageable. Several features that would normally be included in a production housing were represented only in basic form or omitted from the preliminary model. A more detailed industrial design would additionally include:

- optimized ribs around the bearing seats,
- machined reference surfaces and alignment dowels,
- local reinforcement around the cover bolts,
- detailed gasket grooves and sealing lands,
- lifting points and handling features,
- casting or machining allowances,
- oil-flow channels and drainage details,

- and dedicated inspection and maintenance openings.

Accordingly, the existing CAD model should be interpreted as a functional preliminary housing layout rather than a fully production-ready casing. The major load paths, bearing supports, cover interfaces, and overall packaging were established, while secondary manufacturing and service details were simplified.

19 Manufacturing Considerations

The gearbox was designed as a functional mechanical system rather than as a fully production-released product. Nevertheless, the principal manufacturing routes were considered during the design of the shafts, gears, bearing seats, housing, and sealing interfaces.

19.1 Shaft Manufacturing

The three shafts are manufactured primarily by turning from steel bar stock. The principal machining sequence includes facing, center drilling, rough turning, finish turning of bearing and gear seats, machining of shaft shoulders, cutting of DIN 471 retaining-ring grooves, and milling or broaching of the required keyways.

Bearing seats and seal-running surfaces require tighter dimensional and surface finish control than the remaining shaft regions. These surfaces should therefore be finish-machined after the rough turning operations. Shoulder fillets should be produced with the specified radii in order to reduce stress concentration while still providing sufficient clearance for bearing-ring corner radii.

The small pinions are manufactured integrally with the input and intermediate shafts. After the shaft blank is turned, the pinion teeth may be generated by hobbing or shaping. Integral manufacture improves concentricity and eliminates the need for a separate keyed hub at the pinion location. Because the pinions carry high tooth loads, dimensional inspection of runout, tooth thickness, and tooth alignment is important.

19.2 Gear Manufacturing

The larger driven gears are manufactured as separate components and connected to their shafts by parallel keys. A typical manufacturing sequence consists of turning the gear blank, machining the bore and hub faces, cutting the keyway, generating the helical teeth, applying the required heat treatment, and performing final inspection.



Figure 17: Generating a spur gear with a pinion cutter. (Courtesy of Boston Gear Works, Inc.)

DIN 37Cr4 steel was selected for the gears. The final heat-treatment condition should be controlled so that the assumed bending and contact-strength properties used in the calculations are achieved. Where heat treatment produces significant distortion, the bore, reference faces, or tooth flanks may require corrective finishing.

The profile-shifted pinions and gears must be manufactured using the specified normal module, pressure angle, helix angle, and profile-shift coefficients. Incorrect cutter setup would directly alter the center distance, tooth thickness, and load distribution; therefore, the manufacturing drawing should clearly identify these gear parameters.

19.3 Housing Manufacturing

The housing CAD model was intentionally simplified. For low-volume production, the enclosure could be fabricated from welded steel plates and subsequently machined. For higher production quantities, a cast housing would generally be more economical and would permit the integration of ribs, bosses, oil channels, and service features.

Regardless of the selected manufacturing route, the bearing bores should be machined relative to common datums. The bores supporting each shaft should preferably be finished in one setup, or by a line-boring operation, to preserve shaft-axis alignment. The housing joint faces, bearing-cover interfaces, and sealing lands must also be machined sufficiently flat to prevent leakage and misalignment.

The internal bearing-support wall and external bearing bosses require particular attention because their dimensional accuracy directly affects gear-mesh alignment. Machining allowances should be

included around these regions in a production version of the housing.

19.4 Covers, Spacers, and Retaining Features

Bearing covers and spacers are relatively simple turned or milled components. Their axial faces should be flat and parallel because they participate in the positioning and adjustment of the angular-contact bearing pairs. Spacer lengths should be controlled so that the intended bearing clearance or light preload can be established during assembly.

DIN 471 retaining-ring grooves should be machined to the corresponding standard dimensions. Because these grooves form local reductions in shaft section, the groove geometry should be produced accurately and free from burrs or machining damage. Sharp tool marks in these regions may further increase the local fatigue stress.

19.5 Inspection and Quality Control

The most important manufacturing inspections are:

- shaft and gear-seat diameters,
- bearing-seat roundness and coaxiality,
- gear runout and tooth alignment,
- keyway dimensions and location,
- housing bearing-bore alignment,
- shoulder and spacer axial dimensions,
- and seal-running surface finish.

The completed gearbox should be checked for free rotation, gear backlash, bearing adjustment, shaft end play, and lubricant leakage before powered operation. A short no-load run should be performed before full-load testing in order to identify abnormal noise, temperature rise, or misalignment.

20 Assembly Procedure

The gearbox assembly sequence was planned to reduce the risk of damaging the bearings, seals, and gear teeth while preserving shaft alignment and bearing adjustment. The main shaft groups are prepared before installation into the housing.

20.1 Preparation of Shaft Assemblies

The gears, keys, spacers, retaining rings, and bearing inner rings are first installed on their respective shafts. The two small pinions are integral with the input and intermediate shafts and therefore require no separate mounting operation. The larger driven gears are positioned using shaft shoulders, parallel keys, spacers, and DIN 471 retaining rings.

Before assembly, all shaft seats, keyways, retaining-ring grooves, and bearing surfaces should be cleaned and checked for burrs or damage. Bearings should be mounted by applying force only to the ring being fitted. Where necessary, controlled heating of the bearing inner ring may be used to simplify assembly and avoid impact loading.

20.2 Installation into the Housing

The prepared shaft assemblies are placed into the lower housing section in the following order:

1. input shaft assembly,
2. intermediate shaft assembly,
3. output shaft assembly.

The shaft sequence may be adjusted according to the available assembly clearance, but the gear teeth must not be forced into mesh. The shaft assemblies should be lowered into position while maintaining sufficient clearance between the gear flanks.

The bearing outer rings are then seated in the housing bores and internal bearing-support wall. External bearing covers are installed temporarily to hold the assemblies in position during alignment checks.

20.3 Bearing Adjustment

Each shaft uses a back-to-back angular-contact bearing pair. The bearing pairs are adjusted using end covers, spacers, and, where necessary, shims. The adjustment should remove excessive axial play without introducing excessive preload.

After temporary tightening of the bearing covers, each shaft should rotate freely by hand. Axial end play, bearing smoothness, and resistance to rotation should be checked before final fastening. The intermediate-shaft pair requires particular attention because it carries the highest bearing reaction and governs the bearing-life calculation.

20.4 Gear-Mesh Verification

After the three shafts are installed, the following checks should be performed:

- gear backlash,
- tooth-contact pattern,
- axial position of the gears,
- free rotation through several complete revolutions,
- and absence of interference with the housing or covers.

Backlash should be checked at several angular positions to identify excessive runout or misalignment. A marking compound may be applied to the tooth flanks to evaluate the contact pattern. Uneven contact across the face width may indicate shaft misalignment, bearing-seat deformation, or incorrect spacer dimensions.

20.5 Housing Closure and Final Components

Once the shaft and gear checks are complete, the mating housing surfaces are cleaned and sealed using a suitable gasket or liquid sealant. The upper housing section is installed and connected to the lower section using ten $M8 \times 45$ bolts with matching nuts.

The bolts should be tightened gradually in a cross-pattern to distribute the joint compression uniformly. The external bearing covers, radial shaft seals, non-contact intermediate-shaft sealing elements, oil plugs, breather, and oil-level observation unit are then installed.

20.6 Lubricant Filling and Final Inspection

The gearbox is filled with ISO VG 68 oil to the specified level. Before powered operation, the assembly should be rotated manually to distribute the lubricant and confirm free motion. The final inspection should include:

- shaft rotation and end play,
- bearing noise or roughness,
- gear backlash and contact,
- fastener security,

- seal condition,
- and lubricant leakage.

A short no-load commissioning run should be performed before applying the full design load. During this run, temperature, vibration, noise, and oil leakage should be monitored. Any abnormal behaviour should be corrected before full-load operation.

21 Finite Element Verification

A limited finite-element verification was performed for the output shaft. The analysis was intended as a supplementary check of the manually calculated shaft stresses rather than as the primary design method.

The output shaft model was subjected to the transmitted torque and the gear-force components, while the bearing regions were constrained according to their support functions. The mesh was refined locally around the diameter transitions and retaining-ring grooves, where the highest stress gradients were expected.

The analysis showed that the largest stresses occurred locally at the DIN 471 retaining-ring grooves. These values were significantly higher than the nominal shaft stresses because of the sharp local reduction in section and the associated stress concentration. Therefore, the peak FEA stresses were interpreted as highly localized values rather than as representative stresses over the main shaft body.

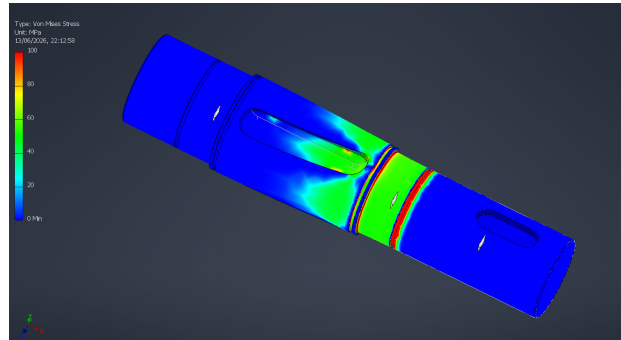


Figure 18: Stress results

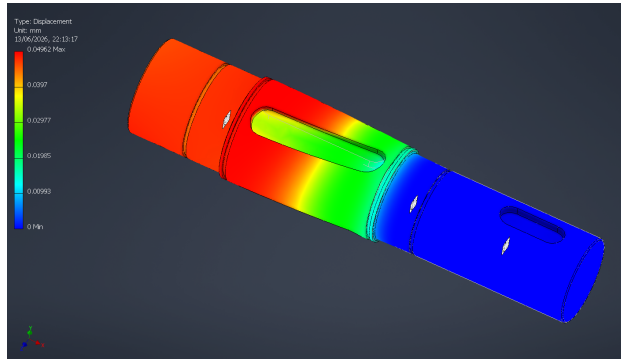


Figure 19: Deformation results

Mesh refinement increased the maximum reported stress from approximately 353 MPa to 380 MPa, corresponding to a change of about 7%. The location of the maximum stress remained at the retaining-ring groove, confirming that this feature governs the local stress field. The manual shaft calculations were therefore retained as the principal strength assessment, while the FEA results were used to identify critical geometric details.

22 Design Limitations and Future Improvements

The present gearbox design satisfies the main functional and strength requirements at the preliminary-design level; however, several aspects were intentionally simplified in order to complete the project within a manageable scope.

The CAD model should not be interpreted as a fully production-ready assembly. In particular, the intermediate-shaft bearing was increased from the preliminary SKF 7309 selection to SKF 7311 BECBP on the basis of the conservative manual reaction calculation. The corresponding shaft seats, housing bores, shoulders, covers, spacers, and retaining features were not updated in the final CAD model. These changes would be required before manufacturing.

The housing geometry was also simplified. A production design would require additional rib optimization, casting or machining allowances, gasket grooves, alignment dowels, lifting features, detailed oil-flow paths, drainage features, and more complete service access.

The dynamic analysis was limited to basic critical-speed checks and a finite element verification of the output shaft. A more advanced study could include:

- full-system modal and harmonic analysis,
- housing and bearing-seat deformation,
- gear-mesh stiffness variation,

- thermal expansion and bearing-preload changes,
- lubricant temperature and churning-loss estimation,
- and prototype testing under representative load cycles.

The selected lubricant, sealing arrangements, bearing clearances, and oil level should therefore be verified experimentally during commissioning. Final manufacturing drawings should also include complete fits, tolerances, surface finishes, heat-treatment specifications, and inspection requirements.

23 Conclusion

A two-stage parallel-shaft helical gearbox was designed for a nominal input power of 15 kW at 1750 rpm. The selected transmission ratio of 20 : 1 reduces the output speed to approximately

$$n_{\text{out}} = 87.5 \text{ rpm.} \quad (348)$$

The gearbox uses two helical gear stages with ratios of 4 : 1 and 5 : 1. Manual gear-strength calculations identified the second-stage gear pair as the governing contact-strength condition. The manual contact safety factor remained slightly above the selected preliminary requirement, whereas Autodesk Inventor returned a lower value and identified this stage as marginal. The pinion tooth-root bending checks satisfied the selected long-life strength criterion.

The shaft calculations included static strength, infinite-life fatigue, deflection, and critical-speed verification. The intermediate shaft was the governing shaft, with a fatigue safety factor of approximately

$$n_f = 1.52. \quad (349)$$

The input and output shafts retained larger safety margins, and all shaft operating speeds remained far below their estimated first critical speeds.

The bearing design was governed by the high radial reaction at the intermediate-shaft B support. The preliminary radial-load-based calculation indicated a baseline life above the 12000 h target after increasing the final analytical selection to SKF 7311 BECBP bearings arranged back-to-back. The existing CAD model retains the earlier preliminary bearing geometry; a production-ready revision would require the corresponding shaft-seat and housing changes. Final manufacturer-specific verification of the combined radial and axial bearing loads also remains necessary.

The final design also incorporates keyed large gears, integral pinions, DIN 471 retaining rings, spacers, radial shaft seals at the external shafts, a non-contact sealing arrangement at the intermediate shaft, ISO VG 68 oil-bath lubrication, and a split housing with an internal bearing-support wall.

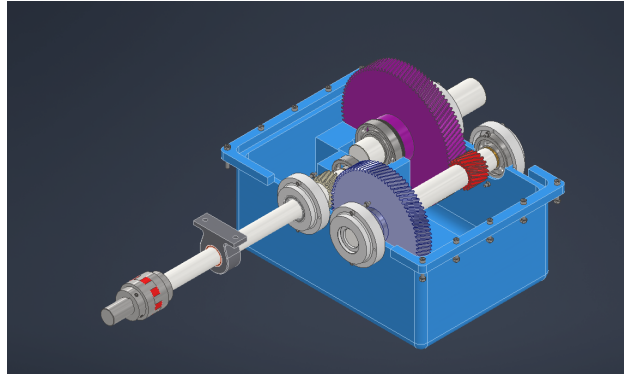


Figure 20: Final Render of the gearbox CAD model, top half is open.

Overall, the gearbox meets the principal performance and strength requirements of the project. The result should be regarded as a validated preliminary mechanical design. Before production, the remaining CAD revisions, detailed manufacturing specifications, and prototype-level thermal, vibration, and lubrication tests should be completed.

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